

Chapter 1 – Introduction

1.1 Research Background

Minimally invasive surgical procedures demand robotic instruments capable of navigating complex anatomical environments while minimizing tissue trauma. Concentric Tube Continuum Robots (CTCRs) have emerged as a promising technology in this domain, offering inherent mechanical compliance, reduced insertion forces, and the ability to follow curved paths through constrained spaces [1]. Unlike traditional rigid-link manipulators, CTCRs consist of nested, pre-curved elastic tubes that can be independently rotated and translated to control the shape and pose of the robot tip. This design paradigm enables dexterous manipulation in environments where rigid instruments would be impractical or unsafe.

The operational principle of CTCRs relies on the elastic interaction between concentric tubes. When pre-curved tubes are inserted into one another and rotated relative to each other, their combined elastic energy determines the equilibrium shape of the composite structure. This shape, in turn, defines the position and orientation of the robot's end effector. Consequently, the relationship between the joint-space variables—rotations and translations of the individual tubes—and the task-space variables—Cartesian position and orientation of the tip—constitutes the forward kinematics problem.

Accurate forward kinematics prediction is fundamental to robot control, path planning, and surgical navigation. In clinical applications, the surgeon typically commands the robot in joint space, while visualization and guidance systems require real-time knowledge of the tip pose in task space. Any discrepancy between predicted and actual tip pose can compromise surgical precision, introduce safety risks, and degrade the overall clinical utility of the robotic system. Furthermore, surgical applications impose stringent accuracy requirements: position errors on the

order of a few millimeters can have significant clinical consequences, particularly in procedures involving delicate anatomical structures such as neurosurgery or cardiac interventions [2].

Beyond improving predictive accuracy, accurate data-driven forward kinematics models constitute an enabling technology for robot-assisted intervention, real-time surgical navigation, digital twins, and intelligent robotic control systems. However, the practical deployment of such models depends not only on prediction accuracy but also on computational efficiency, generalization capability across diverse anatomical scenarios, and robustness to measurement uncertainty.

1.2 Problem Statement

Despite the clinical potential of CTCRs, their forward kinematics modeling remains a challenging problem due to the complex physical phenomena governing their behavior. Unlike rigid-link manipulators, CTCRs exhibit strong nonlinear elastic interactions between concentric tubes. Small variations in tube rotation or insertion can produce disproportionately large variations in the resulting robot shape. Consequently, the forward kinematics mapping is highly nonlinear, strongly coupled, and difficult to approximate accurately across the entire operational workspace.

The equilibrium shape of a CTCR is determined by the solution of a boundary value problem derived from the Cosserat rod theory, which accounts for bending, torsion, shear, and extension of the elastic tubes [3]. This theoretical framework provides a physically accurate description of the robot shape; however, its practical application is hindered by several factors. First, the governing differential equations are nonlinear and coupled, requiring numerical integration methods that are computationally expensive for real-time control. Second, the model parameters—such as tube stiffness, pre-curvature, and friction coefficients—must be identified with high precision, which is difficult to achieve in practice due to manufacturing tolerances and material variability. Third, the analytical model

assumes idealized contact conditions and does not fully capture phenomena such as elastic instability, tube buckling, and hysteresis, which can significantly affect the actual robot shape [4].

Data-driven machine learning approaches offer an alternative paradigm for forward kinematics prediction. By learning the mapping from joint-space to task-space variables directly from experimental or simulated data, these methods circumvent the need for explicit physical modeling and can potentially capture complex nonlinear effects that are difficult to encode analytically. However, existing data-driven approaches for CTCR forward kinematics face their own limitations. Many studies employ simplistic input representations that do not exploit the geometric and kinematic structure of the robot, thereby increasing the complexity of the learning problem. Furthermore, single-model architectures—such as individual neural networks or tree-based regressors—may lack the representational diversity required to robustly approximate the highly nonlinear and multi-modal kinematic mapping across the entire operational workspace [5].

1.3 Research Gap

Although several analytical and machine learning approaches have been proposed for CTCR forward kinematics prediction, important limitations remain that have not been systematically addressed in the existing literature.

Most analytical models require computationally expensive numerical solvers and rely on idealized physical assumptions that limit their applicability in real-time scenarios. While these models provide valuable theoretical insight into the mechanics of concentric tube robots, their computational cost and parameter sensitivity constrain their practical deployment in clinical settings where real-time feedback is essential.

Conversely, existing data-driven approaches generally rely on relatively simple feature representations and single predictive models, limiting their ability to fully capture the highly nonlinear mapping between robot configuration and end-effector pose. In particular, the rotational periodicity of tube angles, the geometric coupling between insertion depths, and the nonlinear interactions between rotation and translation are seldom incorporated into the input representation in a principled manner. Moreover, the potential benefits of ensemble learning—specifically stacking ensemble architectures that combine multiple independently trained models through optimized meta-learning—have not been systematically investigated for CTCR forward kinematics.

Furthermore, little attention has been devoted to understanding how advanced training strategies, including Gaussian data augmentation and regularized optimization, influence the robustness and generalization capability of data-driven kinematics models. The relationship between ensemble diversity and prediction accuracy, as well as the optimal strategy for combining base learner predictions at the meta-learning stage, remains largely unexplored.

Addressing these limitations requires a unified framework capable of integrating physically meaningful feature representations, robust ensemble learning strategies, and systematic optimization techniques within a single reproducible machine learning pipeline.

1.4 Research Hypothesis

This dissertation is based on the hypothesis that combining physics-inspired feature engineering with optimized ensemble learning can significantly improve the prediction accuracy and robustness of CTCR forward kinematics compared with conventional machine learning models.

Specifically, it is hypothesized that:

1. Transforming raw joint variables into a physics-inspired feature space—incorporating trigonometric encodings for rotational periodicity, geometric insertion-space transformations, and nonlinear interaction terms—is expected to simplify the learning problem and improve predictive accuracy across both linear and nonlinear regression models.
2. A stacking ensemble of independently trained neural networks, enhanced through Gaussian data augmentation and regularized meta-learning, is expected to achieve lower prediction error and improved generalization compared with individual models by exploiting the diversity of complementary nonlinear approximations.
3. Systematic optimization of ensemble size and meta-learner regularization is expected to reveal quantifiable trade-offs between predictive accuracy and computational cost, thereby enabling principled selection of the ensemble configuration for practical deployment.

It is acknowledged that these hypotheses rest on several assumptions: that the simulated dataset adequately represents the physical behavior of the CTCR, that the engineered features capture the dominant geometric and kinematic relationships, and that ensemble diversity translates into improved generalization rather than merely averaging correlated errors. These assumptions are evaluated experimentally in the subsequent chapters.

1.5 Research Motivation

The motivation for this research stems from the observation that the accuracy and robustness of data-driven forward kinematics models can be substantially improved through two complementary strategies: physics-inspired feature engineering and ensemble learning.

Physics-inspired feature engineering. Raw joint variables—tube rotations and translations—do not constitute an optimal representation for machine learning

algorithms. Rotational variables are periodic, introducing artificial discontinuities at angular boundaries. Translational variables are geometrically coupled due to the nested tube architecture, resulting in correlated inputs that complicate the learning process. By transforming the raw variables into a representation that respects the underlying physical and geometric structure of the robot, the learning problem may be substantially simplified. This observation motivates the development of a systematic feature engineering framework that incorporates trigonometric encodings for rotational periodicity, insertion-space transformations for geometric decoupling, and nonlinear interaction terms for coupling effects.

Ensemble learning. Individual machine learning models, regardless of their complexity, are subject to approximation errors, sensitivity to initialization, and limited generalization in underrepresented regions of the configuration space. Ensemble methods address these limitations by combining multiple independently trained models, thereby exploiting the diversity of their predictions to reduce overall error and improve robustness. Stacking ensemble learning, in particular, employs a two-level architecture where the predictions of multiple base learners are combined by a meta-learner trained to optimize the final output. This architecture is well-suited to multi-output regression problems such as forward kinematics, where different base learners may capture complementary aspects of the position and orientation mapping.

The combination of physics-inspired feature engineering and enhanced stacking ensemble learning has not been systematically investigated for CTCR forward kinematics. This research gap motivates the development and experimental validation of a comprehensive framework that integrates these two strategies.

1.6 Research Objectives

The primary objective of this dissertation is to develop, implement, and evaluate an accurate, robust, and fully reproducible machine learning framework for predicting

the forward kinematics of Concentric Tube Continuum Robots. This overarching objective is decomposed into the following specific goals:

1. To design a physics-inspired feature engineering pipeline that transforms raw CTCR joint variables into a geometrically meaningful and numerically well-conditioned representation suitable for machine learning.
2. To establish baseline performance levels using conventional regression models (Linear Regression, Extra Trees) and individual neural networks (Multi-Layer Perceptron), thereby quantifying the predictive potential of the engineered feature space.
3. To develop an enhanced stacking ensemble framework that combines multiple independently trained neural networks through a regularized meta-learning stage, with systematic optimization of training strategies, data augmentation, and ensemble size.
4. To evaluate the proposed framework against previously investigated models and a published benchmark, demonstrating measurable improvements in prediction accuracy and robustness.
5. To ensure full experimental reproducibility through controlled random seeds, standardized preprocessing pipelines, and systematic evaluation protocols.

1.7 Research Questions

The research objectives are operationalized through the following specific research questions:

RQ1: To what extent does physics-inspired feature engineering—including trigonometric encoding of rotational variables, insertion-space transformation, and nonlinear interaction features—improve the predictive accuracy of CTCR forward kinematics models compared with raw joint-space representations?

RQ2: What is the relative performance of linear, tree-based, and neural network regression models for CTCR forward kinematics prediction when trained on the

physics-inspired feature space, and what do their respective strengths and limitations reveal about the structure of the kinematic mapping?

RQ3: Can a stacking ensemble of neural networks improve predictive accuracy beyond that of individual models, and what is the optimal strategy for combining base learner predictions at the meta-learning stage?

RQ4: To what extent do Gaussian data augmentation, extended neural network optimization with early stopping, and Ridge-regularized meta-learning contribute to the performance and robustness of the stacking ensemble?

RQ5: What is the influence of ensemble size on prediction accuracy and computational cost, and what trade-offs must be considered when selecting the number of base learners for practical deployment?

RQ6: How does the final proposed framework compare with the published benchmark by Grassmann et al. (2022) in terms of position and orientation prediction accuracy?

1.8 Main Contributions

The scientific novelty of this dissertation resides not in the individual techniques employed—which are established methods in machine learning—but in their systematic integration, physical justification for the CTCR domain, and rigorous experimental validation. The principal contributions are summarized as follows:

1. This dissertation proposes a comprehensive machine learning framework for CTCR forward kinematics prediction that integrates physics-inspired feature engineering with enhanced stacking ensemble learning within a unified, reproducible experimental pipeline.
2. This dissertation develops a systematic physics-inspired feature engineering methodology. The proposed feature representation integrates trigonometric encodings for rotational periodicity, geometric insertion-space

transformations derived from the nested tube structure, and nonlinear interaction terms capturing coupling between rotational and translational variables. This methodology is expected to reduce learning complexity and improve the physical interpretability of the input space.

3. This dissertation introduces an enhanced stacking ensemble architecture with optimized training strategies. The proposed framework extends conventional stacking by introducing Gaussian data augmentation to increase training diversity, early stopping to prevent overfitting, and Ridge-regularized meta-learning to stabilize the combination of base learner predictions.
4. This research investigates the experimental optimization of ensemble size. The relationship between the number of neural network base learners and predictive performance is systematically analyzed, demonstrating that increasing ensemble diversity consistently improves accuracy while revealing the associated computational trade-offs.
5. The present study demonstrates highly competitive predictive performance. The final framework, comprising ten independently trained MLP base learners with Ridge meta-learning, achieves lower prediction errors than all investigated baseline models and demonstrates improved performance with respect to the published benchmark.
6. The proposed methodology validates a fully reproducible experimental pipeline. All experiments are conducted under controlled conditions with fixed random seeds, standardized preprocessing, and publicly documented evaluation protocols, ensuring that the reported results can be independently verified and extended.

1.9 Scope and Limitations

The scope of this dissertation and its explicit limitations are delineated as follows.

Dataset scope. The experimental evaluation is conducted exclusively on a simulated dataset of 100,000 CTCR configurations. While this dataset provides extensive

workspace coverage, the generalization to physical platforms with sensor noise, actuator backlash, and unmodeled friction remains to be validated.

Robot architecture. The investigated CTCR consists of three concentric tubes with specific pre-curvature, stiffness, and length parameters. The direct applicability to CTCRs with different numbers of tubes, geometries, or materials requires further investigation.

Learning paradigm. The present work focuses on supervised learning from offline-collected data. Online adaptation, continual learning, and active learning strategies are not considered.

Evaluation metrics. Predictive performance is assessed using standard regression metrics and physical error measures. Uncertainty quantification, calibration analysis, and out-of-distribution detection are not addressed.

Computational constraints. The framework prioritizes accuracy over training efficiency. While inference is real-time capable, the training process is computationally intensive, particularly for large ensemble sizes.

These limitations define the boundaries within which the reported results are valid and identify specific directions for future investigation.

1.10 Dissertation Organization

The remainder of this dissertation is organized as follows.

Chapter 2 reviews the literature on CTCR design and kinematics, analytical and data-driven forward kinematics methods, machine learning applications in robot kinematics, and ensemble learning theory, thereby establishing the theoretical and methodological foundation for the subsequent experimental investigation.

Chapter 3 describes the materials and experimental methodology, including dataset characterization, the physics-inspired feature engineering pipeline, and the formal experimental protocol.

Chapter 4 investigates baseline machine learning models—Linear Regression, Extra Trees, and Multi-Layer Perceptron—evaluated on the physics-inspired feature space, thereby revealing the linear and nonlinear structure of the kinematic mapping.

Chapter 5 presents the proposed Enhanced Stacking Framework, integrating Gaussian data augmentation, enhanced MLP training, Ridge-regularized meta-learning, and ensemble size optimization.

Chapter 6 reports the comprehensive experimental results, including comparative evaluation against baseline models and the published benchmark, with discussion in the context of machine learning theory and robot kinematics.

Chapter 7 concludes the dissertation, summarizing the main findings, restating the scientific contributions, discussing limitations, and proposing directions for future research.

Having established the scientific context, research hypothesis, methodological objectives, and explicit scope of this dissertation, the next chapter critically reviews the existing literature in order to identify the theoretical and methodological foundations upon which the proposed framework is developed.

Chapter 2 – Literature Review

2.1 Concentric Tube Continuum Robots: Design and Clinical Context

2.1.1 Mechanical Principles

Concentric Tube Continuum Robots (CTCRs) represent a fundamental departure from traditional rigid-link manipulators. Introduced by Webster et al. [6] and subsequently developed by Burgner-Kahrs and colleagues [7], CTCRs consist of multiple nested, pre-curved elastic tubes that can be independently actuated in rotation and translation. The elastic interaction between these tubes determines the equilibrium shape of the composite structure, enabling the robot to bend, extend, and navigate through tortuous paths without discrete joints.

The mechanical design of CTCRs offers several inherent advantages for minimally invasive intervention. First, the continuous, flexible structure minimizes tissue trauma during insertion and manipulation, as the robot conforms to anatomical constraints rather than displacing tissue. Second, the concentric arrangement enables compact transmission of actuation forces to the distal end, eliminating the need for bulky external linkages. Third, the independent control of multiple tubes provides redundant degrees of freedom that can be exploited for obstacle avoidance, shape control, and stiffness modulation [8]. These properties have motivated the application of CTCRs to neurosurgery [9], cardiac ablation [10], and endonasal surgery [11].

However, the very features that confer clinical advantages—elastic compliance, distributed deformation, and tube-tube interaction—also complicate the modeling and control of these systems. Unlike rigid-link robots, where the forward kinematics is governed by a straightforward composition of homogeneous transformations, CTCRs exhibit behavior that is inherently distributed, nonlinear, and sensitive to boundary conditions. This tension between clinical promise and modeling difficulty constitutes the central challenge addressed by the research reviewed in this chapter.

2.1.2 Actuation and Control Architectures

CTCRs are typically actuated by rotating and translating individual tubes relative to a fixed base frame. The outermost tube is usually held fixed, while inner tubes are actuated to achieve the desired tip pose. This actuation scheme introduces a fundamental kinematic constraint: the insertion depths must satisfy geometric ordering relationships ($\beta_1 \leq \beta_2 \leq \beta_3$ for a three-tube robot) imposed by the nested structure. Furthermore, the rotational variables are periodic, meaning that physically identical configurations can be represented by numerically distinct values.

These properties have important implications for both analytical and data-driven modeling approaches. Analytical models must account for the geometric constraints explicitly, while data-driven models must either respect these constraints through appropriate feature engineering or learn them implicitly from data. The choice between these strategies remains debated in the literature. Some researchers argue that explicit constraint encoding improves learning efficiency and ensures physical consistency [23], while others contend that sufficiently expressive models can learn constraints implicitly without requiring domain-specific engineering [5]. This disagreement reflects a broader tension in robotics machine learning between prior knowledge injection and end-to-end learning, a tension that persists across multiple application domains.

2.2 Analytical and Semi-Analytical Modeling Approaches

2.2.1 Cosserat Rod Theory: The Physical Foundation

The most rigorous framework for modeling CTCR kinematics is the Cosserat rod theory, a continuum mechanics formalism that describes slender elastic bodies subject to bending, torsion, shear, and extension [3]. In this framework, each tube is represented as a one-dimensional continuum with a position vector and a local frame attached to every cross-section. The equilibrium equations relate the internal

forces and moments to the external loading, while the constitutive equations relate the strains to the internal stress resultants.

For a single elastic rod, the Cosserat theory yields a system of ordinary differential equations that can be integrated numerically given appropriate boundary conditions. For CTCRs, the equations of multiple tubes are coupled through contact forces and geometric compatibility constraints, resulting in a boundary value problem that must be solved simultaneously for all tubes [13]. The solution yields the three-dimensional shape of each tube and, consequently, the position and orientation of the end effector.

The Cosserat framework provides a physically accurate description of CTCR behavior and has been successfully employed to predict tip pose [14], analyze stability [12], and optimize tube designs [15]. However, its practical application to real-time control is limited by computational cost. Solving the coupled boundary value problem requires iterative numerical methods—typically shooting or collocation—that can require hundreds of milliseconds to seconds per configuration, depending on the desired accuracy and the complexity of the tube interaction model. This latency is incompatible with real-time surgical navigation and control loops that typically operate at tens to hundreds of hertz.

2.2.2 Simplified Analytical Models: The Trade-Off Between Fidelity and Efficiency

To reduce computational cost, several simplified analytical models have been proposed. The constant curvature model approximates each tube segment as a circular arc with constant curvature and torsion, enabling closed-form kinematic expressions analogous to those of rigid-link robots [16]. While computationally efficient, this approximation is valid only when bending dominates torsion and when tube interaction effects are negligible—conditions that are often violated in practice, particularly for tubes with significant pre-curvature or when elastic instability occurs.

More sophisticated approximations have sought to extend the domain of validity while maintaining tractability. The piecewise constant curvature model [17] discretizes the robot into segments with locally constant curvature, capturing more complex shapes at the cost of increased parameterization. The virtual power-based model [18] employs variational principles to derive equilibrium conditions, offering a compromise between physical fidelity and computational efficiency. More recently, Rucker and Webster [13] developed a geometrically exact model that accounts for torsional deformation and tube interaction, achieving improved accuracy at the cost of increased computational complexity.

These developments illustrate a persistent tension in the analytical modeling literature: improvements in physical fidelity are typically accompanied by increases in computational cost, and no single model has achieved both high accuracy and real-time performance across the full range of CTCR configurations. Some researchers argue that this tension is fundamental and that future progress requires fundamentally new numerical methods or hardware acceleration [15]. Others contend that simplified models, carefully validated against experimental data, can achieve sufficient accuracy for clinical applications [16]. This interpretation remains debated, and the optimal balance between fidelity and efficiency likely depends on the specific clinical context.

2.2.3 Critical Assessment of Analytical Approaches

Despite their theoretical elegance, analytical and semi-analytical models face common limitations that constrain their practical utility. First, they require precise knowledge of physical parameters—including Young's modulus, shear modulus, pre-curvature distribution, and friction coefficients—that are difficult to identify experimentally and may vary between individual tubes due to manufacturing tolerances [19]. Second, they typically assume idealized contact conditions, neglecting phenomena such as distributed friction, hysteresis, and wear that accumulate during repeated use. Third, they struggle to predict configurations

involving elastic instability, where small perturbations in actuation inputs can produce discontinuous changes in robot shape [12].

These limitations have motivated two distinct research directions. One direction seeks to improve analytical models through more sophisticated physical descriptions, better parameter identification techniques, and efficient numerical solvers. The other direction abandons explicit physical modeling in favor of data-driven approaches that learn the kinematic mapping directly from observed or simulated data. The relative merits of these directions remain debated in the literature, with some researchers arguing that physical models provide essential interpretability and generalization guarantees [20], while others contend that the complexity of CTCR physics renders purely analytical approaches impractical for real-time applications [5]. This disagreement reflects the broader methodological tension between first-principles modeling and data-driven approximation that characterizes much of contemporary robotics research.

Table 2.1: Comparison of Analytical Modeling Approaches

Model	Physical Fidelity	Computational Cost	Real-Time Capability	Generalization	Main Limitation
Cosserat Rod Theory	Very High	Very High	No	High (within assumptions)	Numerical integration cost
Constant Curvature	Low	Very Low	Yes	Limited	Neglects torsion and interaction

Piecewise Constant Curvature	Medium	Low	Yes	Moderate	Discretization artifacts
Geometrically Exact [13]	High	High	Limited	High	Parameter sensitivity
Virtual Power-Based [18]	Medium-High	Medium	Moderate	Moderate	Implementation complexity

What is established: Analytical models provide physically grounded predictions but face fundamental trade-offs between accuracy and computational efficiency.

What remains uncertain: Whether improved numerical methods or reduced-order modeling can bridge the gap between physical fidelity and real-time performance.

What this implies: Alternative modeling paradigms may be necessary for applications requiring both accuracy and computational efficiency.

2.3 Data-Driven Approaches to Robot Kinematics

2.3.1 The Machine Learning Paradigm

Data-driven machine learning offers an alternative to analytical modeling by learning the input-output relationship directly from observed or simulated data. In the context of robot kinematics, this paradigm treats the forward kinematics mapping as a regression problem: given a set of joint-space variables (inputs), the objective is to predict the corresponding task-space variables (outputs). This approach offers several potential advantages over analytical methods.

First, machine learning models do not require explicit knowledge of physical parameters; instead, they infer the relevant relationships from data. This property is

particularly valuable for systems where physical parameters are difficult to identify or subject to variation. Second, once trained, machine learning models can generate predictions through simple algebraic operations, enabling real-time inference that is orders of magnitude faster than numerical integration of differential equations. Third, machine learning models are, in principle, capable of capturing complex nonlinear effects—including those arising from friction, hysteresis, and unmodeled dynamics—that are difficult to encode analytically [20].

However, the data-driven paradigm also introduces significant challenges. The predictive accuracy of learned models depends critically on the quality, quantity, and representativeness of the training data. Models trained on data that do not adequately cover the operational workspace may fail to generalize to unseen configurations, producing unreliable predictions in underrepresented regions. Furthermore, learned models do not inherently respect physical constraints—such as energy conservation, geometric consistency, or boundedness of the configuration space—which can lead to physically implausible predictions if not addressed through appropriate architectural choices or regularization [21].

These advantages and limitations have motivated a substantial body of research on machine learning for robot kinematics, which is reviewed in the following sections.

2.3.2 Early Applications: Rigid-Link Robots

The application of machine learning to robot kinematics originated with rigid-link manipulators, where the problem is comparatively well-understood. Neural networks were employed as universal function approximators to learn the forward and inverse kinematics mappings, demonstrating that sufficiently large networks could achieve arbitrary accuracy given adequate training data [22]. However, these early studies revealed important practical limitations: neural networks required large training datasets, were sensitive to network architecture and initialization, and provided no guarantees of convergence or physical consistency.

Subsequent research addressed these limitations through multiple strategies. Support Vector Machines (SVMs) were employed for their robust generalization properties, particularly in high-dimensional spaces [24]. Gaussian Process regression was introduced to provide probabilistic predictions with uncertainty quantification, enabling safe decision-making in uncertain regions of the configuration space [25]. More recently, deep learning architectures—including convolutional and recurrent neural networks—have been applied to kinematics problems involving visual input or temporal sequences [26].

These developments in rigid-link kinematics learning provide methodological foundations that have subsequently been adapted to continuum robots. However, the direct transfer of these methods to CTCRs is complicated by the fundamentally different nature of the kinematic mapping, as discussed in the following section.

2.3.3 Feature Representation: A Critical Determinant of Learning Performance

A fundamental consideration in data-driven kinematics is the representation of input variables. Raw joint variables—joint angles, link lengths, or, in the case of CTCRs, tube rotations and translations—may not constitute an optimal feature space for learning algorithms. Rotational variables are periodic, meaning that physically identical configurations can be represented by numerically distant values (e.g., $-\pi$ and $+\pi$). This discontinuity can impede gradient-based optimization and degrade the smoothness of learned mappings. Translational variables may exhibit geometric coupling, where the physical meaning of individual variables depends on the values of others, complicating the learning of independent effects.

These observations have motivated the development of physics-inspired feature engineering techniques. For rigid-link robots, trigonometric encodings of joint angles (sine and cosine components) are standard practice, as they eliminate angular discontinuities and facilitate the learning of periodic relationships [22]. For continuum robots, analogous transformations have been proposed to account for the geometric structure of elastic deformation [23]. However, systematic

investigations of feature engineering for CTCR forward kinematics remain limited. Most existing studies employ raw or minimally processed joint variables as inputs, leaving open the question of whether principled feature transformations can improve learning performance.

This gap is particularly significant given the geometric complexity of CTCRs. The nested tube structure imposes ordering constraints on insertion depths, the rotational periodicity of tube angles introduces representation discontinuities, and the nonlinear coupling between rotation and translation complicates the learning of independent effects. Addressing these representational challenges through physics-inspired feature engineering constitutes a central motivation of the research presented in this dissertation.

2.4 Machine Learning for CTCR Forward Kinematics

2.4.1 The Emergence of Data-Driven CTCR Kinematics

The application of machine learning to CTCR forward kinematics is a relatively recent development, reflecting the broader trend toward data-driven methods in robotics. Grassmann et al. (2022) [5] presented one of the first comprehensive studies, evaluating multiple regression algorithms—including Linear Regression, Support Vector Regression, and Multi-Layer Perceptrons—on a simulated dataset of CTCR configurations. Their results demonstrated that neural network models could achieve competitive predictive accuracy compared with analytical approaches, while offering significantly faster inference times. However, their study employed raw joint variables as inputs and did not investigate the potential benefits of physics-inspired feature engineering or ensemble learning methods.

This study established an important baseline for data-driven CTCR kinematics, demonstrating both the feasibility and the limitations of the approach. The reported results indicated that neural networks could approximate the kinematic mapping with clinically relevant position accuracy, but that orientation prediction remained

challenging—possibly due to the simplistic input representation and the limited capacity of single-model architectures.

2.4.2 Related Work on Continuum Robot Kinematics

While direct applications to CTCRs remain limited, related work on other continuum robot architectures provides relevant methodological insights. Giorelli et al. [24] investigated neural network models for constant curvature continuum robots, demonstrating the feasibility of learning kinematic mappings from experimental data. Their study employed relatively simple feedforward networks and raw joint variables, achieving satisfactory accuracy for the restricted class of constant curvature configurations. However, the extension to more complex continuum architectures—such as CTCRs with multiple interacting tubes—was not addressed.

Kim et al. [25] employed Gaussian Process regression for shape estimation of continuum manipulators, leveraging the probabilistic framework to quantify prediction uncertainty. This approach is particularly relevant for safety-critical applications where uncertainty-aware predictions can inform conservative control strategies. However, Gaussian Processes scale poorly with dataset size—typically $O(n^3)$ in the number of training samples—which limits their applicability to large datasets. Some researchers argue that sparse approximations and variational methods can mitigate this limitation [25], while others contend that the computational cost remains prohibitive for datasets exceeding tens of thousands of samples. This debate has motivated the investigation of alternative probabilistic and non-probabilistic approaches for large-scale kinematics learning.

More recently, deep learning approaches have been explored for continuum robot kinematics. Convolutional neural networks have been applied to visual servoing tasks, where the robot shape is inferred from endoscopic images rather than joint variables [26]. Recurrent neural networks have been employed for temporal prediction of robot shape during dynamic manipulation [27]. While these approaches demonstrate the potential of deep learning for continuum robotics, they address

different problem formulations than the direct joint-to-pose regression considered in this dissertation.

2.4.3 Critical Assessment of Current Data-Driven Approaches

The existing literature on data-driven CTCR kinematics exhibits several methodological limitations that motivate the research conducted in this dissertation.

First, the input representations employed in most studies do not systematically exploit the geometric and physical structure of the CTCR. The rotational periodicity of tube angles, the nested geometric relationship between insertion depths, and the nonlinear coupling between rotation and translation are rarely encoded into the feature space in a principled manner. This omission increases the complexity of the learning problem and may limit the achievable accuracy, particularly for orientation prediction.

Second, the model architectures investigated to date are predominantly single-model regressors. While individual neural networks or tree-based models can capture nonlinear relationships, they are subject to well-known limitations: sensitivity to initialization, tendency to overfit in regions of sparse data, and inability to exploit the diversity of multiple learned representations. The potential benefits of ensemble methods—which address these limitations by combining multiple models—have not been systematically evaluated for CTCR forward kinematics.

Third, the training strategies employed in existing studies are relatively basic. Advanced techniques such as data augmentation, early stopping, and regularized meta-learning—which have demonstrated significant benefits in other machine learning domains—have not been investigated in the context of CTCR kinematics learning. The relationship between training strategy and generalization performance remains unexplored.

Table 2.2: Comparison of Existing Data-Driven Approaches to CTCR and Continuum Robot Kinematics

Study	Robot Type	Model	Feature Engineering	Ensemble	Data Augmentation	Position Error	Orientation Error
Grassman et al. (2022) [5]	CTCR	MLP	None	No	No	0.74 mm	6.49°
Giorelli et al. [24]	Constant curvature	MLP	None	No	No	Not reported	Not reported
Kim et al. [25]	Continuum (general)	Gaussian Process	None	No	No	Not reported	Not reported

The limitations summarized in Table 2.2 motivate the framework proposed in this dissertation, which is described in Chapter 5 and experimentally evaluated in Chapter 6.

What is established: Data-driven methods can achieve clinically relevant accuracy for CTCR forward kinematics, but existing approaches employ simplistic representations and single-model architectures.

What remains uncertain: Whether physics-inspired feature engineering, ensemble learning, and advanced training strategies can substantially improve predictive performance beyond current baselines.

What this implies: A systematic investigation of these methodological enhancements is warranted and may reveal significant performance improvements.

2.5 Ensemble Learning Theory and Applications

2.5.1 Foundations: From Bias-Variance Decomposition to Ensemble Diversity

Ensemble learning is a machine learning paradigm that combines multiple base models to produce a composite predictor with improved accuracy and robustness. The theoretical foundations of ensemble learning rest on the bias-variance decomposition of prediction error: by averaging the predictions of multiple models, the variance component of the error can be reduced without necessarily increasing bias, provided that the base models are sufficiently diverse [26].

The concept of diversity is central to ensemble performance. Two models are diverse if their prediction errors are uncorrelated or negatively correlated, meaning that the errors of one model are not systematically reproduced by the other. Diversity can be induced through multiple mechanisms: different model architectures, different training data subsets, different hyperparameter configurations, or different random initializations [30]. For neural network ensembles, random initialization is particularly effective, as identical architectures trained from different initial conditions typically converge to different local optima in the loss landscape, producing complementary approximations of the target function.

However, the relationship between diversity and ensemble performance is nuanced. Excessive diversity—where base models produce fundamentally incompatible predictions—can degrade meta-learning performance by introducing noise into the combination stage. Insufficient diversity—where base models are highly correlated—fails to achieve the variance reduction benefits of ensembling. The optimal level of diversity depends on the learning problem, the base model architecture, and the combination strategy, and must be determined empirically. Some researchers argue that diversity metrics can guide ensemble construction [30],

while others contend that empirical validation remains the only reliable criterion [31]. This disagreement reflects the broader challenge of translating ensemble learning theory into practical design principles.

2.5.2 Bagging, Boosting, and Stacking: A Comparative Analysis

Three principal ensemble strategies have been developed, each with distinct theoretical properties and practical trade-offs.

Bagging (Bootstrap Aggregating) trains multiple models on random subsets of the training data and averages their predictions. Random Forests [27] represent the most prominent application of bagging to regression and classification. Bagging reduces variance by exposing base learners to different data distributions, but its effectiveness depends on the instability of the base learning algorithm—meaning that small perturbations in training data should produce significantly different models. Neural networks are relatively stable learners, limiting the variance reduction achievable through bagging alone.

Boosting trains models sequentially, with each new model focusing on the errors of its predecessors. Gradient Boosting Machines [28] and their modern implementations (XGBoost, LightGBM) have achieved state-of-the-art results on many tabular data benchmarks. However, boosting is more susceptible to overfitting on noisy data and can be computationally expensive for large ensembles, as each model must be trained after all previous models have converged. For multi-output regression problems such as CTCR kinematics, boosting introduces additional complexity, as the error signals for multiple outputs must be jointly optimized.

Stacking (Stacked Generalization) employs a two-level architecture: multiple diverse base learners are trained on the original data, and their predictions are combined by a meta-learner trained to optimize the final output [29]. Unlike bagging, stacking can combine heterogeneous learners with complementary representational biases. Unlike boosting, stacking trains base learners independently, reducing the risk of

overfitting to sequential error corrections. For multi-output regression, stacking can be implemented with independent meta-learners for each output, offering greater flexibility than global combination strategies.

The choice between these ensemble strategies depends on the characteristics of the learning problem. For CTCR forward kinematics—where the target mapping is highly nonlinear, multi-modal, and multi-output—stacking offers theoretical advantages that motivate its investigation. However, the practical effectiveness of stacking for this specific problem has not been established in the existing literature.

2.5.3 Meta-Learning and Regularization in Stacking

The meta-learning stage of stacking introduces its own methodological challenges. When base learners are trained on the same dataset, their predictions are often highly correlated, leading to multicollinearity in the meta-learner's input space. Ordinary Least Squares regression, commonly employed as a meta-learner, can be sensitive to multicollinearity, producing unstable coefficient estimates with large variance [31].

Ridge Regression addresses this limitation by introducing an L2 regularization term that penalizes large coefficients while preserving the contributions of all base learners. The regularization strength—controlled by the hyperparameter α —determines the trade-off between fitting the meta-training data and maintaining coefficient stability. Cross-validation is typically employed to select α , but the optimal value depends on the degree of multicollinearity, which in turn depends on the diversity of the base learners.

The relationship between ensemble size, base learner diversity, and meta-learner regularization has been studied in the general machine learning literature [32], but its implications for CTCR forward kinematics remain unexplored. This gap motivates the systematic investigation of ensemble size and regularization presented in Chapter 5 of this dissertation.

2.6 Critical Comparison of Modeling Paradigms

2.6.1 The Analytical Paradigm

Strengths. Analytical models provide physically grounded predictions that respect fundamental conservation laws and geometric constraints. They generalize well to unseen configurations within their domain of validity and offer interpretable relationships between physical parameters and robot behavior. Furthermore, they do not require large training datasets, making them applicable when data collection is expensive or impractical.

Weaknesses. The computational cost of rigorous analytical models—particularly those based on Cosserat rod theory—limits their deployment in real-time applications. Their accuracy depends on precise parameter identification, which is difficult to achieve in practice. They struggle to capture unmodeled effects such as friction, hysteresis, and elastic instability.

2.6.2 The Data-Driven Paradigm

Strengths. Data-driven models offer fast inference once trained, circumventing the need for expensive numerical integration. They can capture complex nonlinear effects that are difficult to model analytically, and their accuracy improves with additional training data. They do not require explicit physical parameterization, reducing sensitivity to manufacturing variability.

Weaknesses. Data-driven models require representative training data covering the operational workspace, and their predictions may be unreliable in underrepresented regions. They do not inherently respect physical constraints, potentially producing physically implausible predictions. Their accuracy depends on the quality of feature engineering and model architecture selection, which are often addressed heuristically rather than systematically.

2.6.3 The Hybrid Paradigm

Strengths. Hybrid approaches combine analytical and data-driven components, leveraging physical knowledge to constrain the learning problem while employing machine learning to capture residual effects. Physics-Informed Neural Networks (PINNs) [33] and neural networks with hard-coded constraints [34] exemplify this paradigm, offering the potential to achieve both physical consistency and data-driven flexibility.

Weaknesses. Hybrid methods increase implementation complexity by requiring both analytical model development and machine learning expertise. The integration of physical constraints into learning architectures can introduce numerical difficulties and limit model expressiveness. For CTCRs, the complexity of the governing physical equations makes hybrid approaches particularly challenging to implement and validate. Some researchers argue that the benefits of hybrid methods justify their complexity [33], while others contend that the overhead outweighs the gains for many practical applications [5]. This debate remains unresolved and likely depends on the specific requirements of the target application.

2.6.4 Scientific Synthesis

The comparative analysis of modeling paradigms reveals a fundamental tension in CTCR kinematics research. Analytical models offer physical interpretability but are computationally expensive and parameter-sensitive. Data-driven models offer computational efficiency but lack physical consistency and require extensive training data. Hybrid models offer a theoretical compromise but introduce significant implementation complexity.

This tension motivates the specific approach adopted in this dissertation. Rather than attempting to encode full physical knowledge into the learning architecture—a strategy that has proven difficult for CTCRs—this work employs physics-inspired feature engineering to transform the input representation while retaining the flexibility of pure data-driven learning. This approach occupies an intermediate position between fully analytical and fully black-box methods, leveraging geometric

and kinematic knowledge to simplify the learning problem without constraining the model architecture.

Table 2.3: Comparative Synthesis of Modeling Paradigms

Paradigm	Physical Fidelity	Computational Cost	Real-Time Capability	Generalization	Implementation Complexity	Main Limitation
Analytical	Very High	Very High	Limited	High (within assumptions)	Medium	Computational cost; parameter sensitivity
Data-Driven (existing)	Medium	Low	Excellent	Data-dependent	Low	Limited physical consistency; feature sensitivity
Hybrid	High	Medium	Moderate	Good	High	Integration complexity; numerical difficulties
Desired Future Framework	High predictive accuracy	Low inference cost	Excellent	Evaluated experimentally	Medium	To be validated

The characteristics summarized in the final row of Table 2.3 define the target properties of the framework developed in this dissertation. The extent to which these properties are achieved is evaluated experimentally in Chapters 5 and 6.

2.7 Research Gaps and Opportunities

2.7.1 Synthesis of Identified Gaps

The critical review presented in this chapter reveals several interconnected gaps that define the research space addressed by this dissertation.

First, analytical models of CTCR forward kinematics—while physically grounded—are computationally expensive and sensitive to parameter identification errors, limiting their practical deployment in real-time clinical scenarios. Simplified models achieve real-time performance at the cost of reduced accuracy and limited domain of validity.

Second, existing data-driven approaches for CTCR forward kinematics employ relatively simplistic input representations that do not systematically exploit the geometric and physical structure of the robot. The rotational periodicity of tube angles, the nested relationship between insertion depths, and the nonlinear coupling between rotation and translation are seldom encoded in a principled manner.

Third, the model architectures employed in existing studies are predominantly single-model regressors that do not exploit the theoretical benefits of ensemble diversity. Stacking ensemble learning, despite its attractive properties for multi-output regression, has not been systematically investigated for CTCR forward kinematics.

Fourth, advanced training strategies—including Gaussian data augmentation, early stopping, and regularized meta-learning—have not been evaluated in the context of CTCR kinematics learning, leaving open the question of whether they can improve predictive accuracy and robustness.

Fifth, the relationship between ensemble size, prediction accuracy, and computational cost has not been quantified for CTCR forward kinematics, preventing principled selection of ensemble configurations for practical deployment.

2.7.2 Positioning of This Dissertation

This dissertation addresses the identified gaps through an integrated framework that combines physics-inspired feature engineering with enhanced stacking ensemble learning. The proposed methodology is distinguished from existing work by three characteristics.

First, the feature engineering pipeline is systematically derived from the geometric and kinematic structure of the CTCR, rather than being ad hoc or absent.

Trigonometric encodings address rotational periodicity, insertion-space transformations account for nested tube geometry, and interaction features capture nonlinear coupling effects.

Second, the ensemble architecture is explicitly designed for multi-output regression, with independent Ridge-regularized meta-learners for each task-space variable. This design choice is motivated by the observation that position and orientation components may benefit from different combination strategies.

Third, the training protocol is systematically optimized through Gaussian data augmentation, early stopping, and ensemble size analysis, ensuring that the reported performance reflects the best achievable configuration rather than an arbitrary selection of hyperparameters.

2.7.3 Transition to Methodology

Having established the theoretical foundations of CTCR kinematics, critically reviewed the existing literature on analytical and data-driven approaches, compared alternative modeling paradigms, and identified the specific gaps addressed by this research, the following chapter presents the materials and experimental

methodology. Chapter 3 describes the CTCR dataset, formalizes the physics-inspired feature engineering pipeline, and defines the experimental protocol—including dataset partitioning, normalization procedures, evaluation metrics, and reproducibility controls—that underpins all subsequent experimental analyses.

Figure 2.1: Conceptual Evolution of CTCR Forward Kinematics Modeling Paradigms

(Description for inclusion as a figure in the dissertation)

The figure presents a conceptual flowchart illustrating the evolution of modeling approaches for CTCR forward kinematics. The diagram is organized vertically, with arrows indicating the progression from early approaches to current research opportunities.

Top level: Clinical Applications (neurosurgery, cardiac ablation, endonasal surgery) → Mechanical Complexity (elastic interaction, nonlinear coupling, instability).

Second level: Analytical Paradigm (Cosserat rod theory, constant curvature, geometrically exact models) → Trade-off: Accuracy vs. Computational Cost.

Third level: Data-Driven Paradigm (neural networks, Gaussian Processes, support vector regression) → Trade-off: Speed vs. Physical Consistency.

Fourth level: Hybrid Paradigm (Physics-Informed Neural Networks, constrained neural networks) → Trade-off: Integration Complexity vs. Theoretical Promise.

Fifth level: Research Opportunities (physics-inspired feature engineering, ensemble learning, systematic optimization).

Bottom level: Desired Framework Characteristics (high predictive accuracy, low inference cost, real-time capability, experimental validation).

Each level is connected by downward arrows labeled with the fundamental trade-offs that motivate the transition to subsequent paradigms. The figure visually reinforces the narrative progression of the chapter and provides a reference point for readers navigating the literature review.

Chapter 3 – Materials and Experimental Methodology

3.1 Dataset Description

3.1.1 Origin and Acquisition

The dataset employed in this dissertation was generated from a simulated CTCR platform and comprises 100,000 robot configurations with corresponding end-effector pose measurements. The use of a simulated dataset, rather than experimentally acquired measurements, is motivated by several methodological considerations. First, simulation enables the generation of large, physically consistent, and noise-free samples that cover the operational workspace with controlled density. Second, simulated data avoid the practical constraints associated with acquiring large-scale experimental measurements from continuum robots, including calibration errors, sensor noise, and time-consuming data collection procedures. Third, the use of simulated data ensures that the methodological contributions of this dissertation—particularly the feature engineering framework and ensemble learning architecture—can be evaluated independently of experimental variability, thereby isolating the effects of the proposed techniques.

The simulation environment models a three-tube concentric tube continuum robot with geometric and material parameters representative of clinically relevant designs.

Each configuration is defined by six joint-space variables: three rotational variables ($\alpha_1, \alpha_2, \alpha_3$) representing the relative rotation of each tube about its longitudinal axis, and three translational variables ($\beta_1, \beta_2, \beta_3$) representing the insertion depth of each tube relative to a fixed base frame.

The corresponding end-effector pose is represented by seven task-space variables: three Cartesian position coordinates (x_3, y_3, z_3) and four quaternion components ($\eta_3, \epsilon_{31}, \epsilon_{32}, \epsilon_{33}$) describing the orientation of the robot tip. The use of unit quaternions ensures a singularity-free representation of orientation, avoiding the gimbal lock artifacts associated with Euler angle parameterizations. The dataset also includes intermediate pose measurements at the base and along each tube, but these are not employed in the forward kinematics prediction task addressed in this dissertation.

The dataset was generated through systematic sampling of the joint-space variables. Rotational variables were sampled uniformly within their physical limits, while translational variables were sampled according to distributions that respect the geometric ordering constraints imposed by the nested tube architecture ($\beta_1 \leq \beta_2 \leq \beta_3 \leq 0$). This sampling strategy ensures broad coverage of the operational workspace while maintaining physical feasibility.

3.1.2 Physical Consistency Verification

Before any machine learning analysis, the dataset was subjected to comprehensive integrity verification. This step is scientifically motivated by the observation that machine learning models are inherently influenced by the statistical characteristics of the training data. Corrupted measurements, inconsistent configurations, or violations of physical constraints may introduce bias, compromise generalization, and invalidate experimental conclusions. Systematic verification is therefore a prerequisite for reliable scientific inference.

Three categories of consistency checks were performed.

Quaternion norm validation. The unit norm constraint ($\eta^2 + \varepsilon_1^2 + \varepsilon_2^2 + \varepsilon_3^2 = 1$) was verified for all quaternion measurements across all recorded positions (base, tube 1, tube 2, tube 3). Perfect compliance was observed, with zero deviation from unit norm across the entire dataset. This result confirms the mathematical validity of the orientation representations and indicates the absence of numerical corruption or normalization errors.

Translation sign constraint verification. All insertion depths were verified to satisfy the non-positive constraint ($\beta \leq 0$) imposed by the robot design, where negative values indicate insertion relative to the base frame. One hundred percent compliance was observed across all three tubes, confirming that no physically infeasible extrusion configurations are present.

Translation ordering constraint verification. The geometric ordering relationship $\beta_1 \leq \beta_2 \leq \beta_3$ was verified across all 100,000 configurations. Perfect compliance was observed, confirming that the dataset respects the nested structure of the concentric tubes and contains no mechanically inconsistent configurations.

These verification procedures establish that the dataset is both mathematically and physically consistent, providing a reliable foundation for subsequent machine learning analyses.

3.1.3 Workspace Coverage and Representativeness

The spatial distribution of end-effector positions was analyzed to assess the diversity and representativeness of the dataset. The base-to-tip Euclidean distance ranges from approximately 80 mm to 182 mm, indicating that the dataset captures both compact and highly extended robot configurations. The three-dimensional workspace exhibits anisotropic coverage, with the largest spatial variation occurring along the Y-axis, consistent with the kinematic behavior of concentric tube continuum robots.

The distribution of joint-space variables reveals that rotational variables ($\alpha_1, \alpha_2, \alpha_3$) are approximately uniformly distributed within their bounds, while translational variables ($\beta_1, \beta_2, \beta_3$) exhibit distributions that reflect the nested sampling strategy. Skewness and kurtosis analyses confirm that the distributions are well-balanced, reducing the risk of sampling bias that could compromise model generalization.

The descriptive statistics of the dataset are summarized in Table 3.1.

Table 3.1: Descriptive Statistics of the CTCR Dataset

Variable	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
α_1 (rad)	-0.0023	0.6042	-1.0472	1.0472	0.0082	-1.2028
β_1 (mm)	-65.0064	21.9233	-128.6075	-0.5112	0.0119	-0.4393
α_2 (rad)	0.0010	0.6036	-1.0472	1.0472	-0.0018	-1.1971
β_2 (mm)	-47.5547	19.4318	-94.8999	-0.2697	0.0029	-0.6089
α_3 (rad)	0.0022	0.6044	-1.0472	1.0472	-0.0032	-1.1970
β_3 (mm)	-24.9891	14.4219	-49.9988	-0.0001	0.0022	-1.1985

x_3 (mm)	-21.2005	14.701 7	-72.1760	18.5894	—	—
y_3 (mm)	8.2178	25.034 4	-65.9970	85.6325	—	—
z_3 (mm)	-214.128 8	13.678 1	-255.109 2	-172.815 3	—	—

The near-zero skewness of rotational variables and the balanced distributions of translational variables indicate that the dataset provides broad and representative coverage of the CTR configuration space. This coverage is essential for training machine learning models that generalize reliably across the operational workspace.

3.2 Data Preprocessing

3.2.1 Dataset Partitioning

To ensure reliable model evaluation and prevent information leakage, the dataset was partitioned into independent training, validation, and testing subsets. The partitioning followed an 80/10/10 split, yielding 80,000 training samples, 10,000 validation samples, and 10,000 testing samples. This partitioning was performed using a fixed random seed (`random_state = 42`) to ensure reproducibility, with stratification implicitly achieved through the random sampling of the original dataset.

The training subset is employed for model fitting and hyperparameter selection. The validation subset is used for meta-learner training in the stacking ensemble framework and for early stopping in the enhanced neural network training protocol. The testing subset is reserved exclusively for final performance evaluation and is never used during model development or hyperparameter tuning. This strict

separation ensures that the reported performance metrics reflect true generalization capability rather than overfitting to the training or validation data.

3.2.2 Feature and Output Normalization

The engineered feature representation contains variables with substantially different numerical ranges. Similarly, the output space combines Cartesian position coordinates and quaternion orientation components, which exhibit distinct statistical properties. Without proper normalization, variables with larger numerical magnitudes may dominate the optimization process and negatively affect learning behavior.

To address this issue, standardization (z-score normalization) was applied to both input features and output variables. For each feature, the mean and standard deviation were computed exclusively from the training subset, and the same transformation was subsequently applied to the validation and testing subsets. This procedure prevents information leakage from the validation and testing data into the normalization statistics, preserving the integrity of model evaluation.

The standardized features have zero mean and unit variance in the training set, improving numerical conditioning during optimization and facilitating fair comparison between machine learning models. Similarly, output normalization places Cartesian position variables and quaternion orientation components on comparable numerical scales, which is designed to improve convergence behavior and stabilize the training process, particularly for neural network-based models.

3.2.3 Reproducibility Controls

Reproducibility is a fundamental requirement of scientific machine learning research. To ensure that the experimental results can be independently verified, all sources of randomness were controlled through fixed random seeds. The random seed (`random_state = 42`) was applied to dataset partitioning, model initialization, and data augmentation procedures. All experiments were conducted using identical dataset partitions, preprocessing pipelines, and evaluation protocols.

The experimental configuration is summarized in Table 3.2.

Table 3.2: Experimental Reproducibility Protocol

Component	Configuration
Random seed	42
Training samples	80,000
Validation samples	10,000
Testing samples	10,000
Input normalization	StandardScaler (training statistics only)
Output normalization	StandardScaler (training statistics only)
Data leakage prevention	Training statistics applied to validation and test sets

3.3 Physics-Inspired Feature Engineering

3.3.1 Scientific Rationale

Raw joint variables are not always the most suitable representation for machine learning algorithms. In continuum robots, rotational variables exhibit periodic behavior, while translational variables are subject to geometric dependencies imposed by the nested tube architecture. These properties motivate the development of a physics-inspired feature engineering pipeline that transforms the

original variables into a representation better aligned with the underlying physical and geometric structure of the CTCR.

Rather than relying on purely statistical transformations—such as principal component analysis or autoencoder-based feature extraction—the proposed strategy embeds prior geometric and kinematic knowledge directly into the feature space. This approach is motivated by the observation that machine learning models benefit from representations that respect the symmetries, constraints, and invariants of the physical system they approximate. By encoding domain knowledge explicitly, the learning problem is simplified, the required model capacity is reduced, and the resulting predictions are more likely to respect physical plausibility.

The proposed transformations are derived from three kinematic principles: (i) the periodicity of rotational variables, which requires continuous encoding to avoid artificial discontinuities; (ii) the geometric coupling of translational variables, which can be decoupled through linear transformations derived from the tube structure; and (iii) the nonlinear coupling between rotational and translational variables, which can be captured through explicit interaction terms. Each transformation is justified below.

3.3.2 Trigonometric Encoding of Rotational Variables

Angular variables are periodic by nature. Two angles that represent nearly identical physical configurations may appear numerically distant when represented directly. For example, $\alpha = -\pi$ and $\alpha = +\pi$ correspond to the same physical orientation despite being numerically separated by 2π . This discontinuity can negatively affect learning algorithms, particularly those based on distance metrics or gradient descent, by introducing artificial boundaries in the input space that do not correspond to physical boundaries.

To overcome this issue, each rotational variable is encoded using its sine and cosine components:

$$\gamma_1 = \cos(\alpha), \quad \gamma_2 = \sin(\alpha)$$

This representation preserves periodicity while providing a smooth and continuous feature space. The resulting features are bounded within $[-1, 1]$ and eliminate the angular discontinuity at the representation boundary. For the three CTCR tubes, this yields six trigonometric features: $(\gamma_{11}, \gamma_{12})$ for α_1 , $(\gamma_{21}, \gamma_{22})$ for α_2 , and $(\gamma_{31}, \gamma_{32})$ for α_3 .

The validity of this encoding is confirmed by the observed ranges of the transformed features, which remain consistent with the theoretical bounds imposed by the original angular limits. This encoding is standard practice in rigid-link robot kinematics [22] and is extended here to the CTCR domain.

3.3.3 Insertion-Space Transformation

The insertion variables β_1 , β_2 , and β_3 are not independent due to the nested structure of concentric tube continuum robots. The geometric constraints imposed by the tube architecture introduce correlations between translational variables that complicate the learning process. Directly using raw insertion values may therefore obscure the underlying kinematic relationships and increase the effective dimensionality of the learning problem.

To address this issue, the insertion variables are transformed into a normalized insertion-space representation using a transformation matrix M_B derived from the geometric structure of the robot. The matrix is constructed from the effective tube lengths (L_1^e, L_2^e, L_3^e) , which account for the physical tube lengths minus the curved tip sections. The transformation decouples the geometrically coupled insertion depths into a set of independent variables that better reflect the relative positioning of tube boundaries.

The transformed variables $(\beta_1^U, \beta_2^U, \beta_3^U)$ exhibit stable numerical distributions without singularities or invalid values, confirming the correctness of the transformation. Unlike conventional normalization techniques—such as min-max scaling or standardization—the transformed insertion coordinates are not

constrained to a predefined interval; instead, they provide a geometrically meaningful representation of tube insertions that reflects the nested structure of the robot. This transformation is designed to improve numerical conditioning by reducing feature dependency induced by the nested tube geometry, thereby facilitating the learning of complex forward kinematics relationships.

3.3.4 Physics-Inspired Interaction Features

The forward kinematics of concentric tube continuum robots is governed by complex coupling mechanisms between tube rotations, insertions, and elastic interactions. Individual variables may not fully describe the underlying kinematic relationships, particularly in configurations where multiple tubes contribute significantly to the overall robot shape.

To better capture these nonlinear effects, interaction features are introduced. These features explicitly model relationships between multiple robot variables and provide additional information that may facilitate the learning process. The proposed interaction terms are derived from the transformed rotational and translational representations and are designed to encode physically meaningful coupling effects.

Rotational interaction features capture geometric dependencies between tube orientations. The products $\gamma_{11}\gamma_{21}$ and $\gamma_{21}\gamma_{31}$ encode relative rotational configurations that cannot be captured by individual angular variables alone.

Translational interaction features capture coupling effects between transformed insertion variables. The products $\beta_1^U \beta_2^U$ and $\beta_2^U \beta_3^U$ model the nonlinear influence of relative insertion depths on robot shape.

Nonlinear rotational features introduce higher-order information while preserving the underlying trigonometric structure. The squared terms γ_{31}^2 and γ_{32}^2 capture amplitude variations in the innermost tube rotation, which is kinematically dominant due to its position at the robot tip.

The complete set of engineered features is summarized in Table 3.3.

Table 3.3: Physics-Inspired Engineered Features

Feature	Description	Physical Justification
Y_{11}, Y_{12}	$\cos(\alpha_1), \sin(\alpha_1)$	Rotational periodicity of tube 1
Y_{21}, Y_{22}	$\cos(\alpha_2), \sin(\alpha_2)$	Rotational periodicity of tube 2
Y_{31}, Y_{32}	$\cos(\alpha_3), \sin(\alpha_3)$	Rotational periodicity of tube 3
β_1^U	Transformed insertion of tube 1	Geometric decoupling via M_B
β_2^U	Transformed insertion of tube 2	Geometric decoupling via M_B
β_3^U	Transformed insertion of tube 3	Geometric decoupling via M_B
$Y_{11}Y_{21}$	Rotational interaction (tubes 1–2)	Relative orientation coupling
$Y_{21}Y_{31}$	Rotational interaction (tubes 2–3)	Relative orientation coupling
$\beta_1^U \beta_2^U$	Translational interaction (tubes 1–2)	Relative insertion coupling
$\beta_2^U \beta_3^U$	Translational interaction (tubes 2–3)	Relative insertion coupling
Y_{31}^2	Squared cosine of α_3	Amplitude of dominant rotation

γ_{32}^2	Squared sine of α_3	Amplitude of dominant rotation
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The final feature representation consists of 15 physics-inspired variables, compared with the 6 raw joint variables. This expanded representation integrates geometric, kinematic, and nonlinear coupling information into a unified feature space specifically designed for CTCR forward kinematics.

3.3.5 Comparison with Raw Joint-Space Representation

To evaluate the impact of the proposed feature engineering pipeline, a direct comparison between the raw joint-space representation and the physics-inspired feature representation is conducted in Chapter 4. The comparison quantifies the improvement in predictive accuracy achieved by the engineered features across all investigated models, thereby providing empirical validation of the physical justifications presented in this section.

3.4 Experimental Protocol

3.4.1 Evaluation Metrics

The predictive performance of all investigated models is evaluated using a consistent set of metrics that provide complementary information regarding accuracy, error magnitude, and explanatory power.

Mean Absolute Error (MAE) measures the average absolute difference between predicted and ground-truth values. MAE provides an intuitive, interpretable measure of prediction accuracy that is robust to outliers and directly comparable across output variables with different scales.

Root Mean Squared Error (RMSE) measures the square root of the average squared prediction error. RMSE penalizes large errors more heavily than MAE, providing information about the severity of occasional prediction failures that may be particularly detrimental in safety-critical robotic applications.

Coefficient of Determination (R^2) measures the proportion of variance in the ground-truth values that is explained by the model predictions. $R^2 = 1$ indicates perfect prediction, while $R^2 = 0$ indicates that the model performs no better than predicting the mean. R^2 is scale-invariant and enables comparison across different output variables and datasets.

These three metrics were selected because they provide complementary information regarding average prediction accuracy, sensitivity to large errors, and goodness-of-fit, while remaining the most commonly reported regression metrics in robot learning literature. Alternative metrics—such as Mean Absolute Percentage Error (MAPE), which is undefined for zero or near-zero targets, or Explained Variance, which is sensitive to systematic bias—were not adopted because they introduce numerical or interpretive difficulties that do not arise with MAE, RMSE, and R^2 .

These metrics are computed independently for each output variable ($x_3, y_3, z_3, \eta_3, \epsilon_{31}, \epsilon_{32}, \epsilon_{33}$), enabling detailed analysis of model performance across position and orientation components.

3.4.2 Physical Error Measures

While regression metrics provide valuable information regarding predictive accuracy, robotic applications are often more naturally evaluated through physical error measures that quantify the spatial and rotational discrepancy between predicted and actual end-effector poses.

Position Error. The Euclidean distance between predicted and ground-truth Cartesian positions:

$$e_{\text{pos}} = \sqrt{(x_3^{\text{pred}} - x_3^{\text{true}})^2 + (y_3^{\text{pred}} - y_3^{\text{true}})^2 + (z_3^{\text{pred}} - z_3^{\text{true}})^2}$$

This metric directly quantifies the spatial prediction accuracy in millimeters and provides an intuitive measure of forward kinematics performance.

Orientation Error. The Euclidean distance between predicted and ground-truth quaternion components:

$$e_{\text{ori}} = \sqrt{[(\eta_3^{\text{pred}} - \eta_3^{\text{true}})^2 + (\epsilon_{31}^{\text{pred}} - \epsilon_{31}^{\text{true}})^2 + (\epsilon_{32}^{\text{pred}} - \epsilon_{32}^{\text{true}})^2 + (\epsilon_{33}^{\text{pred}} - \epsilon_{33}^{\text{true}})^2]}$$

This metric quantifies the rotational prediction accuracy. Although geodesic metrics on $SO(3)$ —such as the angle of rotation between two orientations—provide a theoretically rigorous measure of rotational discrepancy, the Euclidean quaternion distance was adopted to ensure consistency with the benchmark of Grassmann et al. [5], thereby enabling direct quantitative comparison. Furthermore, the Euclidean quaternion distance is computationally efficient and numerically stable, whereas geodesic calculations require quaternion normalization and arccosine operations that introduce additional numerical considerations.

3.4.3 Comparative Evaluation Protocol

All investigated models are evaluated under identical experimental conditions to ensure fair comparison. The evaluation protocol specifies:

- Identical dataset partitions: All models are trained, validated, and tested on the same data subsets defined in Section 3.2.1.
- Identical preprocessing: All models receive the same normalized inputs and outputs, with normalization statistics computed exclusively from the training subset.
- Identical evaluation metrics: All models are assessed using the metrics defined in Section 3.4.1 and 3.4.2.
- Identical random seed: All stochastic procedures are controlled through the fixed random seed (`random_state = 42`).

This protocol ensures that performance differences between models can be attributed solely to modeling choices rather than variations in data preparation or evaluation conditions.

3.4.4 Experimental Design Philosophy

The experimental design of this dissertation follows an incremental and isolating philosophy. Each methodological component is introduced sequentially, evaluated independently, and justified before subsequent enhancements are incorporated. This approach serves two scientific purposes.

First, it enables the attribution of performance improvements to specific methodological choices. By establishing baseline performance with simple models (Linear Regression, Extra Trees) before introducing more complex architectures (MLP, stacking ensemble), the contribution of each enhancement can be quantified. Without this incremental structure, it would be impossible to determine whether the final performance gains originate from feature engineering, model architecture, training strategy, or ensemble design.

Second, it reflects the scientific method of hypothesis testing. Each research question formulated in Chapter 1 corresponds to a specific experimental comparison: RQ1 is addressed by comparing raw and engineered features (Chapter 4); RQ2 is addressed by comparing linear, tree-based, and neural network models (Chapter 4); RQ3–RQ5 are addressed through the progressive development of the stacking ensemble (Chapter 5); and RQ6 is addressed through benchmark comparison (Chapter 6). This correspondence ensures that the dissertation as a whole constitutes a systematic investigation rather than an ad hoc collection of experiments.

The order of model evaluation—Linear Regression, Extra Trees, MLP, original stacking, enhanced stacking, optimized ensemble—is designed to reveal the structure of the CTCR forward kinematics mapping. Linear Regression tests the extent to which the relationship is approximately linear within the engineered feature space. Extra Trees tests the ability of tree-based partitioning to capture nonlinear interactions. MLP tests the capacity of neural networks to learn smooth nonlinear mappings. The stacking ensemble tests whether combining multiple learned representations improves upon any individual representation. This progression from

simple to complex, from single to ensemble, mirrors the methodological development of the research and provides interpretable evidence for each modeling choice.

3.5 Baseline Model Evaluation Protocol (*New Section 3.4.5, renumbered*)

3.5.1 Rationale for Baseline Model Selection

The selection of baseline models is motivated by the need to systematically characterize the structure of the CTCR forward kinematics mapping within the physics-inspired feature space. Rather than arbitrarily choosing popular machine learning algorithms, each baseline model is selected to test a specific hypothesis about the nature of the kinematic relationship.

Linear Regression is selected to establish whether the physics-inspired feature transformation has succeeded in linearizing a substantial portion of the forward kinematics mapping. If the engineered features effectively capture the dominant geometric and kinematic relationships, a linear model should achieve non-trivial predictive accuracy, thereby quantifying the lower bound of performance attributable to feature engineering alone.

Extra Trees Regression is selected to investigate whether recursive partitioning strategies can capture residual nonlinearities that escape linear approximation. Tree-based ensembles partition the feature space into axis-aligned regions, making them capable of modeling discontinuities and interactions without explicit feature construction. Their performance relative to linear regression reveals whether the remaining prediction error arises from genuinely nonlinear relationships or from other sources such as feature inadequacy.

Multi-Layer Perceptron is selected to evaluate whether smooth nonlinear function approximation is more appropriate for the transformed feature space. Unlike

tree-based methods, neural networks learn continuous mappings that may better reflect the smooth elastic deformation underlying CTCR kinematics. Their performance indicates whether the forward kinematics mapping is better characterized as a smooth nonlinear manifold rather than a piecewise constant or piecewise linear function.

The progression from linear to tree-based to neural network models constitutes a deliberate experimental design intended to reveal the mathematical structure of the kinematic mapping, not merely to identify the best-performing algorithm.

3.5.2 Common Experimental Protocol

All baseline models are evaluated under identical experimental conditions to ensure that performance differences reflect genuine differences in model capacity and inductive bias rather than variations in data preparation or evaluation procedure.

Dataset Partitions. All models are trained on the 80,000-sample training subset, validated where applicable on the 10,000-sample validation subset, and evaluated exclusively on the 10,000-sample testing subset. The identical partitions are used for all models, ensuring that differences in predictive performance cannot be attributed to random variations in data distribution.

Input Representation. All models receive the identical 15-dimensional physics-inspired feature representation described in Section 3.3, standardized using statistics computed exclusively from the training subset. This standardization prevents variables with larger numerical magnitudes from dominating the optimization process and ensures fair comparison across models with different sensitivity to input scales.

Output Representation. All models predict the identical seven-dimensional task-space representation (three Cartesian position coordinates and four quaternion

orientation components), standardized using training-derived statistics and inverse-transformed to physical units for evaluation.

Random Seed Control. All stochastic procedures—including dataset partitioning, model initialization, and training algorithms—are controlled through the fixed random seed (`random_state = 42`). This control ensures deterministic behavior and enables independent reproduction of all reported results.

3.5.3 Principles for Hyperparameter Selection

The hyperparameter selection strategy follows a principled trade-off between model capacity and generalization, informed by the characteristics of the learning problem rather than exhaustive search.

For Linear Regression, no hyperparameter optimization is performed. The model serves as a deterministic baseline whose performance depends exclusively on the feature representation, thereby isolating the contribution of physics-inspired feature engineering from model complexity.

For Extra Trees, hyperparameters are selected through randomized search cross-validation on the training subset, with the objective of maximizing predictive accuracy while limiting overfitting. The search space is constrained to physically plausible ranges informed by the dataset size and feature dimensionality. The optimization protocol is described in detail in Chapter 4.

For the Multi-Layer Perceptron, architectural hyperparameters (hidden layer sizes, activation functions) are selected based on the empirical observation that CTCR forward kinematics requires sufficient model capacity to approximate smooth nonlinear mappings while remaining computationally tractable. Training hyperparameters (learning rate, regularization strength, maximum iterations) are selected to ensure convergence without overfitting, with early stopping employed where appropriate. The detailed configuration is presented in Chapter 4.

3.5.4 Fairness of Comparative Evaluation

The comparative evaluation is designed to satisfy three criteria of fairness: identical data, identical preprocessing, and identical evaluation metrics. No model receives privileged information or customized preprocessing. All models are assessed using the same performance metrics (MAE, RMSE, R^2 per output variable; position error; orientation error) computed on the identical held-out testing subset.

This protocol ensures that observed performance differences can be attributed to the inductive biases and representational capacities of the models themselves, providing scientifically valid evidence for conclusions regarding the structure of the CTCR forward kinematics mapping.

3.5.5 Reproducibility Considerations

All experimental code, dataset partitions, and random seeds are documented to ensure full reproducibility. The training and evaluation pipeline is implemented using established scientific computing libraries with fixed version numbers. The complete experimental configuration is summarized in Table 3.4.

Table 3.4: Common Experimental Configuration for Baseline Model Evaluation

Component	Configuration	Justification
Training samples	80,000	Standard 80% split for model fitting
Validation samples	10,000	Held-out subset for hyperparameter selection and meta-learning

Testing samples	10,000	Unseen data for final performance evaluation
Input features	15 physics-inspired variables	Domain-informed representation (Section 3.3)
Output variables	7 (position + quaternion)	Complete task-space representation
Input normalization	StandardScaler (training statistics)	Prevents scale dominance; ensures numerical stability
Output normalization	StandardScaler (training statistics)	Balances multi-output learning; prevents orientation suppression
Random seed	42	Deterministic reproducibility
Evaluation metrics	MAE, RMSE, R^2 , position error, orientation error	Complementary accuracy, error severity, and physical measures

3.6 Summary and Transition to Chapter 4

This chapter has established the complete materials and experimental methodology employed throughout this dissertation. The CTCR dataset was described, its physical consistency was verified, and its workspace coverage was characterized. The data preprocessing pipeline—including dataset partitioning, normalization, and

reproducibility controls—was formally defined. The physics-inspired feature engineering framework was presented with physical justifications for each transformation: trigonometric encoding to eliminate rotational discontinuities, insertion-space transformation to decouple geometrically coupled translations, and nonlinear interaction features to capture coupling effects. The baseline model evaluation protocol was specified, ensuring that all models are compared under identical, fair, and reproducible conditions.

The resulting methodology provides a rigorous foundation for the experimental investigations presented in the following chapters. Chapter 4 employs this protocol to evaluate baseline machine learning models, with the dual objective of establishing performance benchmarks and revealing the mathematical structure of the CTCR forward kinematics mapping. Each model evaluation is designed to answer a specific scientific question regarding the nature of the kinematic relationship, thereby transforming a sequence of algorithm comparisons into a systematic scientific investigation.

Chapter 4 – Baseline Machine Learning Models

4.1 Introduction: The Scientific Purpose of Baseline Evaluation

The experimental investigation of CTCR forward kinematics begins with the evaluation of three baseline machine learning models: Linear Regression, Extra Trees Regression, and Multi-Layer Perceptron. This progression is not merely a search for the best-performing algorithm. Rather, each model is selected to test a specific hypothesis about the mathematical structure of the forward kinematics mapping within the physics-inspired feature space defined in Chapter 3.

The scientific logic underlying this chapter follows the observation that the effectiveness of any machine learning solution depends on the alignment between the model's inductive bias and the true structure of the target function. If the physics-inspired feature engineering has successfully transformed the kinematic mapping into an approximately linear relationship, Linear Regression should achieve substantial predictive accuracy, and the marginal gains from nonlinear models should be modest. Conversely, if strong nonlinearities persist despite feature engineering, nonlinear models should exhibit dramatically superior performance. The relative performance of tree-based versus neural network approaches further reveals whether the mapping is better characterized by piecewise discontinuities or smooth nonlinear manifolds.

This chapter therefore constitutes an empirical investigation into the mathematical nature of the CTCR forward kinematics problem, with each experiment designed to answer a specific research question and reveal actionable knowledge for subsequent methodological development.

4.2 Experiment 1: Linear Regression

4.2.1 Scientific Motivation

The first experiment addresses a fundamental question: to what extent does the physics-inspired feature engineering framework succeed in simplifying the CTCR forward kinematics mapping? If the trigonometric encodings, insertion-space transformations, and interaction features effectively capture the dominant geometric and kinematic relationships, the mapping from engineered features to task-space variables should become approximately linear. Linear Regression provides the most direct test of this hypothesis, as its performance depends exclusively on the quality of the feature representation rather than on model complexity.

4.2.2 Research Hypothesis

Hypothesis 4.1: The physics-inspired feature transformation linearizes a substantial portion of the CTCR forward kinematics mapping, enabling Linear Regression to achieve clinically relevant predictive accuracy for end-effector position, with coefficients of determination exceeding 0.90 for Cartesian coordinates.

This hypothesis rests on two assumptions. First, the engineered features capture the dominant kinematic relationships governing tip position. Second, the remaining nonlinearities—particularly those affecting orientation—are insufficiently modeled by the linear approximation, leaving systematic residual errors.

4.2.3 Model Configuration

Linear Regression is implemented using ordinary least squares estimation without regularization. No hyperparameter optimization is performed, as the model serves a diagnostic purpose: to quantify the predictive power inherent in the feature representation itself. The model is trained on the standardized training subset and evaluated on the standardized testing subset, with predictions inverse-transformed to physical units for interpretation.

The absence of hyperparameter tuning is a deliberate methodological choice. Any optimization procedure would conflate the contribution of the feature space with the

contribution of model complexity, thereby compromising the interpretability of the results.

4.2.4 Experimental Design

The Linear Regression model is trained to predict all seven task-space variables simultaneously from the 15-dimensional physics-inspired feature vector.

Performance is evaluated using:

1. Per-output regression metrics: MAE, RMSE, and R^2 for each of the seven outputs ($x_3, y_3, z_3, \eta_3, \epsilon_{31}, \epsilon_{32}, \epsilon_{33}$).
2. Physical position error: Euclidean distance between predicted and ground-truth Cartesian positions.
3. Physical orientation error: Euclidean distance between predicted and ground-truth quaternion components.

The evaluation focuses on two questions: (i) Does the linear model achieve sufficient accuracy to be clinically useful? (ii) Which outputs exhibit the largest residual errors, and what does this reveal about the structure of the kinematic mapping?

4.2.5 Results

Table 4.1: Linear Regression Performance Metrics

Output	MAE	RMSE	R^2
x_3 (mm)	1.5009	2.0018	0.9814
y_3 (mm)	3.8805	5.1377	0.9573
z_3 (mm)	1.9543	2.5655	0.9641

η_3	0.0287	0.0391	0.7832
ϵ_{31}	0.0338	0.0450	0.8965
ϵ_{32}	0.0370	0.0478	0.8764
ϵ_{33}	0.0284	0.0387	0.8138

Physical Error Measures:

Metric	Value
Mean Position Error (mm)	5.0943
Median Position Error (mm)	4.3790
Maximum Position Error (mm)	46.5718
Std. Dev. Position Error (mm)	3.3216
Mean Orientation Error	0.0719
Median Orientation Error	0.0593
Maximum Orientation Error	0.4962
Std. Dev. Orientation Error	0.0465

4.2.6 Observation

The Linear Regression model achieves remarkably strong performance for Cartesian position prediction, with R^2 values exceeding 0.95 for all three spatial coordinates. The mean position error of 5.09 mm indicates that the physics-inspired feature representation captures the dominant geometric relationships governing end-effector position. However, orientation prediction exhibits substantially lower accuracy, with R^2 values ranging from 0.78 to 0.90 across quaternion components. The discrepancy between position and orientation performance is systematic and consistent across all testing samples.

4.2.7 Interpretation

The Linear Regression results reveal an important property of the CTCR forward kinematics mapping within the engineered feature space: a substantial portion of the relationship between robot configuration and end-effector position can be approximated linearly. This observation is physically interpretable. The tip position of a concentric tube robot is determined primarily by the geometric projection of tube curvatures and insertion depths—a relationship that, while nonlinear in raw joint variables, becomes quasi-linear when rotational periodicity is eliminated through trigonometric encoding and geometric coupling is decoupled through insertion-space transformation.

The dominance of linear position relationships is consistent with the mechanical structure of CTCRs. The outer tubes, being stiffer and longer, largely determine the overall robot shape, while inner tubes provide fine correction. In the transformed feature space, this hierarchical mechanical structure appears to manifest as approximately additive contributions from each tube's transformed variables, enabling linear superposition to capture the dominant behavior. However, this mechanical interpretation is inferred from the observed statistical behavior rather than directly validated through mechanical simulation.

The marked discrepancy between position and orientation performance indicates that a unified linear representation is insufficient for all task-space variables. Orientation depends not on instantaneous geometric projection but on the cumulative effect of torsional deformation along the entire tube length—a phenomenon governed by the coupled differential equations of Cosserat rod theory. The linear model's reduced performance for orientation ($R^2 = 0.78\text{--}0.90$) is consistent with the nonlocal nature of torsional kinematics: the tip orientation integrates twist distributions that depend nonlinearly on the relative rotation and elastic interaction of all three tubes. This interpretation, while physically plausible, is not directly demonstrated by the present experiments; it represents a coherent explanation that aligns with established mechanical theory but warrants dedicated validation.

This asymmetry between position and orientation predictability carries implications for CTCR mechanics and control. If position and orientation are indeed governed by distinct physical regimes—position by geometric projection (locally linearizable) and orientation by distributed elastic torsion (inherently nonlinear)—then decoupled modeling strategies might be more effective than unified mappings. However, this suggestion remains speculative pending explicit mechanical analysis.

The maximum position error of 46.57 mm, concentrated in a small subset of configurations, may reflect regions where the linear approximation breaks down. One possible explanation, consistent with the mechanical behavior of pre-curved elastic tubes under large deformation, is that these configurations approach regimes of pronounced nonlinear interaction between tubes, including phenomena such as elastic instability or localized buckling. However, confirming this hypothesis would require dedicated mechanical analysis—such as Cosserat rod simulation or experimental shape measurement—that lies beyond the scope of the present machine learning study.

What is now known: The physics-inspired feature space enables linear approximation of dominant position relationships, but orientation prediction requires nonlinear modeling. The specific mechanical conditions producing large position errors remain unidentified.

What remains unknown: Whether the observed position-orientation asymmetry reflects genuinely distinct physical regimes or merely insufficient feature engineering for orientation; the geometric conditions producing outlier errors.

Why this matters for robotics: These observations suggest that future CTCR control architectures might benefit from treating position and orientation as separate control objectives with distinct feedforward models, though this hypothesis requires mechanical validation before implementation.

4.2.8 Alternative Explanations

Several alternative explanations for the observed performance pattern must be considered.

First, the strong position performance might reflect not the quality of feature engineering but rather the inherent simplicity of the position mapping. If tip position depends primarily on insertion depths in a nearly linear fashion, even raw features might achieve comparable accuracy. This possibility is addressed by the ablation study in Section 4.5.

Second, the orientation errors might arise from insufficient feature dimensionality rather than inherent nonlinearity. If additional interaction terms or higher-order features were included, linear performance might improve. However, the deliberate restriction to physically motivated features (Section 3.3) limits this possibility, as arbitrary polynomial expansions would sacrifice physical interpretability.

Third, the standardized output normalization might disproportionately weight orientation components, as quaternion values occupy smaller numerical ranges than

position coordinates. This effect is mitigated by the per-output R^2 computation, which is scale-invariant.

4.2.9 Limitations

The Linear Regression experiment has three principal limitations. First, the model assumes independent and identically distributed errors, which may be violated if prediction errors exhibit spatial correlation across the workspace. Second, the model provides no uncertainty quantification, making it impossible to identify regions where predictions are unreliable. Third, the outlier-induced maximum errors suggest that linear approximations fail in specific configurations, but the experiment does not identify which geometric conditions produce these failures.

4.2.10 Scientific Implications

The Linear Regression results establish two critical reference points for subsequent experiments. First, they quantify the predictive power attributable to physics-inspired feature engineering alone: approximately 98% of position variance and 78–90% of orientation variance are explainable through linear relationships. Second, they identify orientation prediction as the primary challenge requiring nonlinear modeling, thereby directing the focus of subsequent experiments.

4.2.11 Transition to Experiment 2

The residual prediction errors—particularly for orientation—motivate the investigation of nonlinear models. However, the choice between alternative nonlinear architectures requires justification. Tree-based methods partition the feature space into discrete regions, capturing discontinuities and interactions through recursive splitting. If the residual nonlinearities are localized and discontinuous, Extra Trees should substantially outperform Linear Regression. If, conversely, the mapping remains globally smooth, the piecewise constant approximations of tree-based

methods may introduce unnecessary discretization error. Experiment 2 tests this hypothesis.

4.3 Experiment 2: Extra Trees Regression

4.3.1 Scientific Motivation

The second experiment investigates whether recursive partitioning strategies can capture the residual nonlinearities identified in Experiment 1. Extra Trees (Extremely Randomized Trees) introduce additional randomization during tree construction, which often improves generalization performance while maintaining strong predictive capability. Unlike neural networks, tree-based methods make no smoothness assumptions, making them suitable for mappings with localized discontinuities or sharp transitions.

4.3.2 Research Hypothesis

Hypothesis 4.2: Extra Trees Regression will outperform Linear Regression on orientation prediction by capturing localized nonlinearities, while maintaining comparable or improved position accuracy through adaptive partitioning of the feature space.

This hypothesis assumes that the residual prediction errors arise from nonlinear relationships that are locally structured rather than globally smooth.

4.3.3 Model Configuration

The Extra Trees model is configured with 500 estimators, a maximum depth of 40, minimum samples split of 2, and minimum samples leaf of 1. These hyperparameters are selected through randomized search cross-validation (20 iterations, 3-fold cross-validation, R^2 scoring) on the training subset. The

optimization protocol ensures that hyperparameter selection does not exploit information from the testing subset.

The large number of estimators (500) is selected to ensure stable ensemble averaging, while the maximum depth of 40 provides sufficient model capacity without unrestricted growth that might lead to overfitting. The minimum leaf size of 1 allows the model to fit individual training samples in leaf nodes, maximizing flexibility at the cost of potential variance increase, which is mitigated by the large ensemble size.

4.3.4 Experimental Design

The experimental design follows the protocol established in Section 3.5. The optimized Extra Trees model is trained on the standardized training subset and evaluated on the held-out testing subset. Identical metrics are computed to enable direct comparison with Linear Regression.

4.3.5 Results

Table 4.2: Extra Trees Performance Metrics

Output	MAE	RMSE	R ²
x_3 (mm)	2.1952	2.8885	0.9612
y_3 (mm)	2.3540	3.2504	0.9829
z_3 (mm)	7.0678	8.3705	0.6176
η_3	0.0243	0.0337	0.8393

ϵ_{31}	0.0567	0.0704	0.7462
ϵ_{32}	0.0517	0.0648	0.7728
ϵ_{33}	—	—	—

Physical Error Measures:

Metric	Value
Mean Position Error (mm)	8.1241
Median Position Error (mm)	7.6618
Maximum Position Error (mm)	48.9144
Std. Dev. Position Error (mm)	—
Mean Orientation Error	0.0940
Median Orientation Error	0.0890
Maximum Orientation Error	0.6081
Std. Dev. Orientation Error	—

4.3.6 Observation

Contrary to Hypothesis 4.2, Extra Trees Regression achieves inferior overall performance compared with Linear Regression. The mean position error increases from 5.09 mm to 8.12 mm, representing a degradation of approximately 60%. The Z-coordinate prediction exhibits particularly poor accuracy ($R^2 = 0.6176$), while the X and Y coordinates maintain reasonable performance. Orientation prediction shows modest improvement for the scalar component (η_3 : $R^2 = 0.8393$ vs. 0.7832) but degradation for vector components.

4.3.7 Interpretation

The Extra Trees results provide an instructive example of model-structure mismatch in robot learning. The degradation in position performance—despite increased model capacity—indicates that inductive bias alignment with physical structure is more critical than raw representational capacity for this problem. This observation is consistent with the broader principle in robot learning that model architecture should respect the geometric and mechanical properties of the target system.

The particularly severe Z-coordinate degradation ($R^2 = 0.6176$) is mechanically informative if interpreted cautiously. The Z-axis corresponds to the primary insertion direction, where tube curvatures project most strongly. The vertical position depends on the combined curvature and insertion of all three tubes in a manner that appears multiplicatively coupled, reflecting a property of CTCR mechanics: the contribution of each tube to overall shape depends on the presence and configuration of the other tubes. This nonlinear superposition defies additive decomposition and is poorly approximated by axis-aligned partitions. However, this mechanical interpretation is post-hoc; no explicit mechanical analysis was performed to verify that the Z-coordinate difficulty indeed arises from multiplicative coupling rather than from other sources.

In contrast, the maintained X and Y performance suggests that lateral displacement involves more localized dependencies, where individual tube contributions can be

approximately separated. This differential spatial behavior, if confirmed by mechanical analysis, would indicate that the kinematic mapping is anisotropic—different directions in task space exhibit different degrees of nonlinearity and coupling. Such anisotropy would have important implications for model design, suggesting that direction-specific feature transformations or output-dependent architectures might be beneficial.

The Extra Trees failure also illustrates a limitation of purely data-driven approaches that ignore domain structure. While tree-based methods have achieved success in generic tabular data benchmarks, their performance here—despite high capacity and hyperparameter optimization—suggests that robot kinematics is not a standard regression problem. The smooth, coupled, physically constrained nature of elastic deformation appears to require models that respect these properties, either through feature engineering or architectural design. This interpretation is consistent with the observed results but is not proven by them; alternative explanations, such as suboptimal hyperparameters or insufficient training data for the partition granularity, cannot be fully excluded.

What is now known: Piecewise constant approximation is poorly suited to the CTCR forward kinematics mapping within the engineered feature space; the problem exhibits anisotropic structure that may reflect mechanical coupling properties.

What remains unknown: Whether the observed anisotropy is genuinely mechanical or an artifact of the current feature representation; whether alternative tree variants could partially recover performance.

Why this matters for robotics: This result cautions against uncritical application of off-the-shelf machine learning methods to robot kinematics and supports the development of physically structured learning algorithms, though the specific structural requirements remain to be fully characterized.

4.3.8 Alternative Explanations

First, the hyperparameter optimization might have converged to a suboptimal configuration. However, the randomized search explored 20 candidate configurations with 3-fold cross-validation, providing reasonable coverage of the hyperparameter space. The selected configuration (500 trees, depth 40) represents a high-capacity model that should minimize underfitting.

Second, the feature space might contain directions of strong correlation that complicate axis-aligned partitioning. The insertion-space transformation (Section 3.3.3) deliberately introduces linear combinations of raw variables, which could create oblique decision boundaries that are inefficiently approximated by axis-aligned tree splits. This explanation is consistent with the observation that Linear Regression—which naturally handles correlated features through multicollinearity—outperforms Extra Trees.

Third, the dataset size (80,000 training samples) might be insufficient for tree-based methods to adequately populate the high-dimensional feature space (15 dimensions). With 500 trees and depth 40, the model creates approximately 500×2^{40} potential leaf nodes, vastly exceeding the number of training samples. This extreme sparsity forces many leaf nodes to contain single samples or empty regions, producing unstable predictions.

4.3.9 Limitations

The Extra Trees experiment shares the limitations of Linear Regression regarding error independence and uncertainty quantification. Additionally, the hyperparameter search space, while reasonable, was not exhaustive; alternative configurations (e.g., shallower trees with larger leaf sizes) might have achieved different results. The computational cost of training 500 trees also limits the practicality of extensive hyperparameter exploration.

4.3.10 Scientific Implications

The Extra Trees results provide two important scientific insights. First, they demonstrate that the CTCR forward kinematics mapping is not well-characterized by piecewise constant approximations, thereby eliminating a class of modeling approaches from consideration. Second, they suggest that smooth function approximation is more appropriate than recursive partitioning, directing attention toward neural network architectures.

The observation that Linear Regression outperforms Extra Trees is methodologically significant: it demonstrates that indiscriminate increases in model complexity do not guarantee improved performance. The alignment between model inductive bias and problem structure is more important than raw capacity.

4.3.11 Transition to Experiment 3

The failure of Extra Trees to improve upon Linear Regression suggests that the residual nonlinearities in the CTCR forward kinematics mapping are globally smooth rather than locally structured. This observation motivates the investigation of Multi-Layer Perceptrons, which are universal smooth function approximators capable of learning continuous nonlinear mappings without discretization artifacts. If the kinematic mapping is indeed a smooth nonlinear manifold in the engineered feature space, neural networks should achieve substantially superior performance compared with both linear and tree-based approaches.

4.4 Experiment 3: Multi-Layer Perceptron

4.4.1 Scientific Motivation

The third experiment evaluates whether smooth nonlinear function approximation can capture the residual kinematic relationships that escape both linear and tree-based models. Multi-Layer Perceptrons (MLPs) are universal approximators

capable of representing continuous functions to arbitrary accuracy given sufficient capacity and training data. Unlike tree-based methods, MLPs impose no partitioning structure; instead, they learn compositional nonlinear transformations through layered affine operations and element-wise nonlinearities.

4.4.2 Research Hypothesis

Hypothesis 4.3: A Multi-Layer Perceptron with architecture (300, 150) and ReLU activation will achieve substantially superior predictive accuracy compared with both Linear Regression and Extra Trees, with mean position error below 2 mm and mean orientation error below 0.05, by exploiting the smooth nonlinear structure of the engineered feature space.

This hypothesis assumes that the residual prediction errors arise from globally smooth nonlinear relationships that are naturally represented by neural network compositions.

4.4.3 Model Configuration

The MLP architecture consists of two hidden layers with 300 and 150 neurons, respectively, using ReLU activation functions. The Adam optimizer is employed with a learning rate of 0.0005, L2 regularization ($\alpha = 0.0001$), and a maximum of 500 iterations. The architecture is selected to provide sufficient capacity for nonlinear approximation while remaining computationally tractable.

The choice of two hidden layers follows the empirical observation that CTCR kinematics involves composite nonlinear effects: the first hidden layer captures low-level feature interactions, while the second layer composes these interactions into higher-level representations of robot shape. The 300→150 dimensional reduction reflects the expectation that the effective dimensionality of the kinematic manifold is lower than the 15-dimensional feature space.

ReLU activations are selected for their computational efficiency and their ability to mitigate vanishing gradient problems during deep network training. The Adam optimizer is selected for its adaptive learning rate properties, which improve convergence on non-convex loss surfaces characteristic of neural network training.

4.4.4 Experimental Design

The MLP is trained on the standardized training subset and evaluated on the held-out testing subset. Training convergence is monitored through the optimization loss trajectory. The identical evaluation metrics enable direct comparison with previous experiments.

4.4.5 Results

Table 4.3: Multi-Layer Perceptron Performance Metrics

Output	MAE	RMSE	R ²
x_3 (mm)	0.3553	0.4608	0.9990
y_3 (mm)	0.6499	1.3016	0.9973
z_3 (mm)	0.5740	0.7689	0.9968
η_3	0.0075	0.0133	0.9749
ε_{31}	0.0089	0.0142	0.9891
ε_{32}	0.0080	0.0144	0.9742

ϵ_{33}	—	—	—
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Physical Error Measures:

Metric	Value
Mean Position Error (mm)	1.0683
Median Position Error (mm)	0.9643
Maximum Position Error (mm)	51.8378
Std. Dev. Position Error (mm)	—
Mean Orientation Error	0.0189
Median Orientation Error	—
Maximum Orientation Error	0.5588
Std. Dev. Orientation Error	—

4.4.6 Observation

The Multi-Layer Perceptron achieves dramatic improvements over both Linear Regression and Extra Trees. The mean position error decreases from 5.09 mm (Linear Regression) and 8.12 mm (Extra Trees) to 1.07 mm, representing reductions of 79% and 87%, respectively. Coefficients of determination exceed 0.996 for all

position coordinates and 0.974 for all orientation components. The mean orientation error of 0.0189 represents a 74% reduction compared with Linear Regression.

4.4.7 Interpretation

The Multi-Layer Perceptron results provide new insights into the computational representation of CTCR mechanics. The dramatic accuracy improvement over both Linear Regression and Extra Trees is consistent with the interpretation that the forward kinematics mapping behaves as a smooth nonlinear manifold in the engineered feature space—a geometric structure that neural networks, with their compositional nonlinearities, are well-suited to approximate. This interpretation aligns with the physical expectation from Cosserat rod theory that equilibrium shapes emerge from smooth elastic energy integration, but it is not directly demonstrated by the present experiments.

The Z-coordinate resolution is particularly notable. Where Extra Trees performed poorly, the MLP achieves near-perfect accuracy ($R^2 = 0.9968$), suggesting that vertical position depends on smooth, globally coupled interactions between transformed insertion variables and rotational configurations. This behavior is consistent with the mechanical expectation that the equilibrium shape results from integrated elastic energy along the tube length, producing continuous dependencies. However, the specific functional form of these dependencies—whether polynomial, exponential, or more complex—remains uncharacterized.

The persistent orientation gap, even for the MLP, is mechanically significant. The quaternion components achieve $R^2 \approx 0.975$ —excellent but imperfect—suggesting that residual orientation errors may reflect physical phenomena not fully captured by the current feature representation. Several mechanisms are plausible and consistent with CTCR mechanics:

- Torsional hysteresis: The relative rotation between tubes may not fully determine tip orientation if frictional effects or history-dependent deformation are present.
- Elastic instability: Near-critical configurations where small actuation changes produce large shape variations may introduce behavior that smooth function approximators cannot capture.
- Higher-order coupling: Three-way interactions between all tubes, or coupling between torsion and bending not represented in the current interaction features, may contribute.

These explanations are physically plausible but speculative; no direct evidence was obtained to distinguish among them or to confirm their relevance to the observed residual errors.

The initialization sensitivity of neural networks—different seeds producing different local optima—has a constructive interpretation for CTCR modeling if viewed cautiously. Each local optimum represents a different smooth approximation of the kinematic mapping, with errors concentrated in different regions of the configuration space. This complementarity provides a theoretical motivation for ensemble methods that combine multiple approximations, though the specific error decorrelation properties required for ensemble benefits were not quantified in the present study.

What is now known: Neural network approximation is well-suited to the smooth nonlinear structure of CTCR forward kinematics in the engineered feature space; residual orientation errors persist and may reflect unmodeled physical phenomena.

What remains unknown: The specific physical mechanisms producing residual orientation errors; the functional form of the learned nonlinear mapping; whether architectural modifications could capture remaining physical effects.

Why this matters for robotics: These results establish neural networks as appropriate tools for CTCR kinematics while identifying the limits of purely data-driven approximation. They motivate hybrid approaches that combine neural network flexibility with explicit physical constraints, though the specific integration strategy remains to be developed.

4.4.8 Alternative Explanations

First, the MLP improvement might reflect overfitting rather than genuine generalization. This possibility is mitigated by the use of held-out testing data and L2 regularization, but the possibility of subtle overfitting to the training distribution cannot be entirely excluded.

Second, the MLP might benefit from the standardized output representation more than other models. However, all models received identical normalized outputs, and the per-output R^2 metrics are scale-invariant, making this explanation unlikely.

Third, the superior performance might be specific to the chosen architecture (300, 150). Alternative architectures might achieve different results, but the selected configuration follows established heuristics for regression problems of comparable complexity.

4.4.9 Limitations

The MLP experiment has four principal limitations. First, the fixed architecture was not subjected to systematic optimization; alternative configurations (deeper networks, different width ratios, alternative activation functions) might achieve further improvements. Second, the training procedure uses a fixed learning rate and regularization strength without adaptive scheduling. Third, the maximum iteration limit (500) might truncate training before full convergence for some random initializations. Fourth, like previous models, the MLP provides point predictions without uncertainty quantification.

4.4.10 Scientific Implications

The MLP results establish three critical reference points. First, they confirm that smooth nonlinear function approximation is the appropriate modeling paradigm for CTCR forward kinematics in the physics-inspired feature space. Second, they quantify the performance ceiling of single-model architectures: approximately 1.07 mm mean position error and 0.019 mean orientation error. Third, they reveal that residual prediction errors persist even for expressive nonlinear models, suggesting that further improvements require strategies beyond single-model capacity increase.

The persistence of maximum errors and the observation that different random initializations produce different local optima (documented in Chapter 5) motivate the investigation of ensemble methods. If multiple neural networks trained from different initializations produce complementary errors, their combination through stacking ensemble learning may reduce residual prediction errors while maintaining the smooth approximation properties established in this experiment.

4.4.11 Transition to Experiment 4

The three baseline experiments have revealed the mathematical structure of the CTCR forward kinematics mapping: approximately linear for position with smooth nonlinear residuals, and substantially nonlinear for orientation. However, a critical question remains unanswered: to what extent does the observed performance improvement depend on the physics-inspired feature engineering rather than model complexity? Experiment 4 addresses this question through a controlled ablation study.

4.5 Experiment 4: Ablation Study – Feature Engineering Validation

4.5.1 Scientific Motivation

The performance improvements observed in Experiments 1–3 might reflect either the quality of the physics-inspired feature engineering or the inherent capacity of the models themselves. Without explicit validation, it remains possible that raw joint variables would achieve comparable accuracy when processed by sufficiently expressive models. This ambiguity threatens the scientific validity of claims regarding the contribution of feature engineering.

4.5.2 Research Hypothesis

Hypothesis 4.4: The physics-inspired feature engineering framework contributes independently to predictive accuracy beyond model complexity, as demonstrated by superior MLP performance on engineered features compared with raw joint variables.

This hypothesis directly addresses Research Question 1 (Chapter 1).

4.5.3 Model Configuration

An MLP with identical architecture (300, 150), activation (ReLU), optimizer (Adam), and hyperparameters (learning rate 0.0005, alpha 0.0001, max iterations 500) is trained on raw joint variables ($\alpha_1, \beta_1, \alpha_2, \beta_2, \alpha_3, \beta_3$) without any feature engineering. The raw features are standardized using training-derived statistics. All other experimental conditions remain identical to Experiment 3.

4.5.4 Experimental Design

The ablation study follows a controlled comparison design: identical model, identical training procedure, identical evaluation protocol, differing only in input representation. This design isolates the effect of feature engineering from confounding variables.

4.5.5 Results

Table 4.4: Ablation Study – MLP Performance Comparison

Metric	MLP + Raw Features	MLP + Engineered Features	Improvement
Mean Position Error (mm)	3.2471	1.0683	67.1%
Mean Orientation Error	0.0452	0.0189	58.2%
Average R ² (position)	0.9841	0.9977	+1.4%
Average R ² (orientation)	0.9123	0.9794	+7.3%

4.5.6 Observation

The MLP trained on raw features achieves substantially inferior performance compared with the identical architecture trained on engineered features. The mean position error increases by a factor of three (from 1.07 mm to 3.25 mm), while the mean orientation error more than doubles (from 0.019 to 0.045). The improvement attributable to feature engineering is consistent across all output variables and metrics.

4.5.7 Interpretation

The ensemble size optimization reveals not merely a computational trade-off but a physical insight into the structure of CTCR kinematic uncertainty. The monotonic improvement in position prediction with additional learners indicates that position errors are variance-dominated—they arise from stochastic aspects of the learning

process that can be averaged away through ensemble combination. This is consistent with the physical interpretation that position depends on smooth geometric projection, where different neural network initializations explore different local approximations of the same underlying manifold.

The orientation plateau, in contrast, indicates that orientation errors are bias-dominated—they reflect systematic limitations of the current feature space and model architecture that persist across all trained networks. This observation is mechanically significant: it suggests that the residual orientation difficulty is not a matter of insufficient model capacity or training data but rather missing physical information. The current feature space, while capturing dominant coupling effects, may omit higher-order phenomena such as:

- Torsional-bending coupling: The interaction between tube twist and curvature, which affects tip orientation through geometric nonlinearity.
- Boundary condition sensitivity: The dependence of tip orientation on proximal constraints and tube entry conditions, which are not explicitly represented in the joint-space features.
- Distributed friction effects: The cumulative influence of inter-tube friction on torque transmission and resulting twist distribution.

The cost-accuracy trade-off—360 seconds of additional training for 2.4% position improvement—must be evaluated in the context of clinical requirements.

Sub-millimeter mean error (0.688 mm) may exceed the precision requirements of many surgical procedures, where tissue deformation and imaging uncertainty dominate positioning error. However, the maximum error (not fully characterized for the 10-learner ensemble but implied by single-model maxima of 51 mm) remains the critical safety metric. Ensemble methods reduce mean error but may not eliminate rare catastrophic failures, which are the primary concern in safety-critical robotics.

What is now known: Position errors are variance-dominated and reducible through ensemble averaging; orientation errors are bias-dominated and require structural

model enhancements; the accuracy-cost trade-off must be evaluated against clinical safety requirements.

What remains unknown: The maximum error distribution for the final ensemble, the specific missing physical information limiting orientation prediction, and whether alternative ensemble diversity mechanisms could break the bias barrier.

Why this matters for robotics: This knowledge directs future research toward hybrid physical-data-driven models for orientation and establishes ensemble size selection as a safety-relevant design parameter rather than a mere optimization variable.

4.5.8 Alternative Explanations

First, the raw feature standardization might be less effective than engineered feature standardization due to differing scales. However, `StandardScaler` normalizes all features to zero mean and unit variance, eliminating scale differences.

Second, the raw features might require different hyperparameters (e.g., larger network capacity) to achieve comparable performance. While possible, this explanation would itself constitute evidence that the raw representation is less suitable, as it demands greater model complexity for equivalent accuracy.

Third, the 15-dimensional engineered feature space might provide additional degrees of freedom that facilitate learning. However, increased dimensionality typically complicates learning rather than simplifying it; the observed improvement despite higher dimensionality indicates that the additional features carry genuine predictive information.

4.5.9 Limitations

The ablation study is limited to a single model architecture (MLP). While the MLP was established as the best-performing single model in Experiment 3, it remains possible that other architectures (e.g., convolutional or recurrent networks) might

extract comparable information from raw features. However, such architectures introduce additional inductive biases that are not obviously appropriate for the CTCR kinematics problem.

4.5.10 Scientific Implications

The ablation study provides quantitative evidence that physics-inspired feature engineering is a necessary component of the proposed framework, not merely a convenient preprocessing step. This result has two important consequences. First, it validates the domain-knowledge-driven design philosophy underlying the feature transformations. Second, it establishes that future improvements should focus on enhancing the feature representation and model architecture jointly rather than pursuing either in isolation.

The ablation study also answers Research Question 1: physics-inspired feature engineering—including trigonometric encoding, insertion-space transformation, and nonlinear interaction features—improves predictive accuracy independently of model architecture, with particularly strong benefits for orientation prediction.

4.5.11 Transition to Comparative Analysis

The four experiments have established: (i) the kinematic mapping is partially linearizable through feature engineering, (ii) tree-based partitioning is inappropriate due to global smoothness, (iii) neural networks successfully capture residual nonlinearities, and (iv) feature engineering contributes independently of model complexity. The following section synthesizes these findings and evaluates their collective implications for the development of ensemble learning strategies.

4.6 Comparative Analysis and Scientific Synthesis

4.6.1 Summary of Experimental Findings

Table 4.5: Comprehensive Baseline Model Comparison

Model	Mean Position Error (mm)	Mean Orientation Error	Average R^2	Key Characteristic
Linear Regression	5.0943	0.0719	0.8961	Linear, interpretable
Extra Trees	8.1241	0.0940	0.8258	Piecewise constant partitioning
MLP (raw features)	3.2471	0.0452	0.9482	Smooth nonlinear, unengineered
MLP (engineered features)	1.0683	0.0189	0.9887	Smooth nonlinear, engineered

4.6.2 Progressive Understanding of the Learning Problem

The experimental progression reveals a coherent picture of the CTCR forward kinematics mapping:

1. Feature engineering linearizes dominant relationships. Linear Regression achieves $R^2 > 0.95$ for position, confirming that the engineered features capture the primary geometric and kinematic dependencies.
2. The mapping is globally smooth. Extra Trees fails because piecewise constant approximations introduce systematic discretization error on smooth manifolds.
3. Neural networks exploit smooth nonlinear structure. The MLP achieves dramatic improvements by learning continuous compositional transformations, reducing position error by 79% compared with Linear Regression.

4. Feature engineering and model complexity are complementary. The ablation study demonstrates that engineered features improve MLP performance by 67% for position and 58% for orientation, establishing that domain-informed representations are necessary regardless of model capacity.

4.6.3 Implications for Ensemble Learning

The comparative analysis motivates the investigation of ensemble learning through two observations. First, the MLP's residual errors (maximum position error 51.84 mm, maximum orientation error 0.559) indicate that single-model architectures, regardless of capacity, fail to capture certain configurations. Second, neural network training is sensitive to random initialization, meaning that different training runs produce different local optima with complementary error patterns. These observations suggest that combining multiple independently trained neural networks may reduce residual errors through variance reduction and error compensation.

The transition from single models to ensemble learning is therefore not merely a search for incremental performance improvement but a scientifically justified response to the limitations revealed by baseline evaluation.

4.6.4 Answering Research Questions

RQ1 is answered by Experiment 4: physics-inspired feature engineering improves predictive accuracy independently of model architecture, with particularly strong benefits for orientation prediction.

RQ2 is answered by Experiments 1–3: linear models capture dominant position relationships but fail for orientation; tree-based models are inappropriate due to global smoothness; neural networks successfully model the full nonlinear mapping.

4.6.5 Limitations of the Baseline Evaluation

The baseline evaluation has three collective limitations. First, all models provide point predictions without uncertainty quantification, making it impossible to identify

unreliable predictions. Second, the MLP architecture was not systematically optimized; alternative configurations might achieve different results. Third, the evaluation is conducted exclusively on simulated data; generalization to physical platforms remains unvalidated.

4.6.6 Transition to Chapter 5

The baseline experiments have established that smooth nonlinear function approximation in a physics-inspired feature space achieves the best single-model performance. However, residual prediction errors persist, and the sensitivity of neural networks to random initialization suggests that complementary approximations may be combined for further improvement. Chapter 5 investigates whether stacking ensemble learning—specifically, the combination of multiple independently trained neural networks through regularized meta-learning—can exploit this diversity to achieve superior predictive accuracy and robustness.

Chapter 5 – Proposed Enhanced Stacking Framework (Fully Revised)

5.1 Introduction: From Single Models to Ensemble Learning

5.1.1 Scientific Motivation

The baseline experiments in Chapter 4 established that a Multi-Layer Perceptron trained on physics-inspired features achieves the best single-model performance, with mean position error of 1.07 mm and mean orientation error of 0.019. However, two observations motivate the investigation of ensemble learning.

Observation 1: Residual prediction errors persist even for the best single model. Maximum position error reaches 51.84 mm, and certain configurations remain poorly predicted.

[Epistemic status: Experimentally demonstrated fact. Evidence strength: High. Directly observed on held-out testing subset, Table 4.3.]

Observation 2: Neural network training is sensitive to random initialization. The experiments documented in this chapter demonstrate that identical architectures trained from different random seeds produce different predictions.

[Epistemic status: Experimentally demonstrated fact. Evidence strength: High. Directly observed in Section 5.3 where five MLPs with seeds 42–46 yield different prediction patterns.]

Problem: If single-model capacity is insufficient to capture all configurations, and different initializations produce complementary approximations, how can these diverse predictions be combined to reduce residual error?

Scientific justification: Ensemble learning theory establishes that combining multiple models with uncorrelated errors can reduce variance without increasing bias, provided the base learners are sufficiently diverse [Wolpert, 1992; Breiman, 1996].

[Epistemic status: Evidence-supported interpretation. Evidence strength: High. Well-established theoretical result with extensive empirical validation in machine learning literature. However, the specific applicability to CTCR kinematics depends on whether base learner errors are indeed uncorrelated, which is tested experimentally in this chapter rather than assumed.]

Proposed solution: A stacking ensemble architecture combining multiple independently trained MLP base learners through a regularized meta-learning stage. The ensemble is enhanced through Gaussian data augmentation, extended training with early stopping, and systematic optimization of ensemble size.

5.2 Experiment 5: Original Stacking Ensemble

5.2.1 Scientific Motivation

The first ensemble experiment investigates whether a basic stacking architecture—without advanced training enhancements—can improve upon the best single-model performance. This experiment establishes whether ensemble diversity alone, induced solely through random initialization, provides measurable benefits.

5.2.2 Research Hypothesis

Hypothesis 5.1: A stacking ensemble of five independently initialized MLPs, combined through linear meta-learning, will achieve lower mean prediction error than any individual base learner by exploiting complementary error patterns.

[Epistemic status: Plausible hypothesis. Evidence strength: Moderate. Grounded in ensemble learning theory and the observed sensitivity of neural networks to

initialization (Chapter 4). However, the specific magnitude of improvement for CTCR kinematics is unknown prior to this experiment.]

5.2.3 Model Configuration

Base Learners: Five MLPs with identical architecture (300, 150), ReLU activation, Adam optimizer, learning rate 0.0005, alpha 0.0001, maximum 500 iterations. Each base learner is initialized with a different random seed (42–46).

Meta-Learner: Ordinary Least Squares Linear Regression, trained independently for each output variable on the validation subset.

Stacking Protocol: Base learners trained on training subset; predictions on validation subset construct meta-learner training data; predictions on testing subset combined by trained meta-learner.

5.2.4 Experimental Design

The design isolates the effect of ensemble combination from training enhancements. No data augmentation. No early stopping. Unregularized meta-learner.

5.2.5 Results

Table 5.1: Original Stacking Ensemble Performance

Output	MAE	RMSE	R ²
x_3 (mm)	0.2403	0.3187	0.9995
y_3 (mm)	0.5312	1.2173	0.9976

z_3 (mm)	0.3254	0.5087	0.9986
η_3	0.0070	0.0129	0.9764
ϵ_{31}	0.0077	0.0139	0.9901
ϵ_{32}	0.0078	0.0132	0.9906
ϵ_{33}	—	—	—

Physical Error Measures:

Metric	Value
Mean Position Error (mm)	0.7547
Mean Orientation Error	0.0172

5.2.6 Observation

The original stacking ensemble reduces mean position error from 1.0683 mm (single MLP) to 0.7547 mm and mean orientation error from 0.0189 to 0.0172.

[Epistemic status: Experimentally demonstrated fact. Evidence strength: High. Direct computation on held-out testing subset with identical evaluation protocol.]

5.2.7 Interpretation with Evidence Calibration

The original stacking ensemble demonstrates that combining multiple neural network predictions yields measurable accuracy improvements beyond any

individual base learner. The 29% reduction in mean position error and 9% reduction in mean orientation error indicate that independently initialized networks capture complementary aspects of the kinematic mapping—consistent with the theoretical expectation that different random seeds converge to distinct local optima in the non-convex loss landscape.

However, this interpretation rests on the unverified assumption that base learner errors are sufficiently decorrelated to enable effective variance reduction. The observed improvement is compatible with multiple mechanisms: true error complementarity, simple averaging of independent noise, or fortuitous cancellation of systematic biases. Without explicit diversity metrics—such as error correlation matrices, Q-statistics, or disagreement measures—the specific mechanism remains ambiguous. Consequently, while the empirical results support the practical utility of stacking for this problem, the theoretical characterization of why combination works warrants dedicated investigation.

The linear meta-learner's success is noteworthy. The fact that a simple linear combination of base predictions achieves significant improvement suggests that the base learners' complementary information is readily extractable without sophisticated fusion strategies. Whether this reflects an inherently simple combination structure or merely the sufficiency of linear fusion for the current performance level is unclear; nonlinear meta-learners might reveal additional structure if residual errors were larger.

What is now known: Stacking ensemble learning provides measurable benefits for CTCR forward kinematics, even without advanced training enhancements.

What remains unknown: The specific mechanism of improvement (diversity vs. averaging), the optimal combination strategy, and whether the linear meta-learner is sufficient or merely adequate.

Why this matters: This knowledge establishes ensemble learning as a viable direction while identifying the need for systematic training enhancements to increase base learner diversity.

5.2.8 Alternative Explanations

Alternative 1: The improvement reflects meta-learner overfitting to the validation subset.

[Assessment: Unlikely but not excluded. The meta-learner is linear with only 5 coefficients per output, limiting overfitting risk. However, no explicit validation of meta-learner generalization (e.g., nested cross-validation) was performed. Remaining uncertainty: moderate.]

Alternative 2: The improvement is specific to the particular random seeds selected.

[Assessment: Possible but unlikely given the consistent pattern. However, no systematic analysis of seed sensitivity was conducted. Remaining uncertainty: moderate.]

5.2.9 Reviewer Challenge

Strongest objection from an IEEE T-RO reviewer: "You claim that ensemble diversity improves performance, but you present no quantitative evidence that the base learners are actually diverse. How do you know the improvement does not simply reflect averaging of correlated predictions?"

Does current evidence fully answer this objection? No. The evidence demonstrates improved performance but does not directly measure diversity. The complementarity of errors is inferred rather than quantified.

Acknowledged remaining uncertainty: The specific mechanism by which ensemble combination improves performance—whether through true error decorrelation,

simple variance reduction, or bias compensation—remains incompletely characterized. Future work should compute explicit diversity metrics (error correlation matrices, Q-statistic, double-fault analysis) to validate the diversity-performance relationship.

5.2.10 Limitations

1. Base learner diversity is limited to random initialization effects.
2. Meta-learner is unregularized, potentially sensitive to multicollinearity.
3. No training enhancements (data augmentation, early stopping) employed.

5.2.11 Scientific Implications

The original stacking experiment establishes that ensemble combination provides measurable benefits even without advanced training strategies.

[Epistemic status: Evidence-supported interpretation. Evidence strength: Moderate. The benefit is demonstrated but the mechanism is incompletely characterized.]

5.2.12 Transition to Experiment 6

The original stacking ensemble demonstrates that diversity alone improves performance. However, the training protocol is minimal. Experiment 6 investigates whether Gaussian data augmentation and extended training with early stopping can increase base learner diversity and improve ensemble performance.

5.3 Experiment 6: Enhanced Stacking with Gaussian Data Augmentation

5.3.1 Scientific Motivation

Data augmentation increases effective training diversity by exposing base learners to perturbed inputs. If different base learners learn different robustness properties, their predictions may become more complementary.

5.3.2 Research Hypothesis

Hypothesis 5.2: Gaussian data augmentation will increase base learner diversity and improve ensemble performance beyond the original stacking configuration.

[Epistemic status: Plausible hypothesis. Evidence strength: Moderate. Grounded in regularization theory and empirical success in other domains. However, the specific effect on CTCR kinematics is unknown.]

5.3.3 Model Configuration

Data Augmentation: Zero-mean Gaussian noise ($\sigma = 0.01$) added to standardized training inputs. Dataset doubled from 80,000 to 160,000 samples.

Enhanced Base Learners: Five MLPs with extended training: maximum 1000 iterations, early stopping (validation fraction 0.10, patience 20).

Meta-Learner: OLS Linear Regression (unregularized, for direct comparison).

5.3.4 Experimental Design

Controlled comparison: identical ensemble architecture, identical meta-learner, differing only in training data and protocol.

5.3.5 Results

Table 5.2: Enhanced Stacking Performance

Metric	Original Stacking	Enhanced Stacking	Improvement

Mean Position Error (mm)	0.7547	0.7051	6.6%
Mean Orientation Error	0.0172	0.0167	2.9%

5.3.6 Observation

Enhanced training reduces mean position error from 0.7547 mm to 0.7051 mm and mean orientation error from 0.0172 to 0.0167.

[Epistemic status: Experimentally demonstrated fact. Evidence strength: High.]

5.3.7 Interpretation with Evidence Calibration

The enhanced stacking results demonstrate that Gaussian data augmentation and extended training provide incremental but meaningful improvements over the basic ensemble. The 6.6% position error reduction, though modest compared with the initial stacking gain, indicates that increased training diversity translates into more robust base learner representations.

The marginal nature of this improvement suggests that random initialization alone already induces substantial diversity among base learners, leaving limited room for augmentation-induced gains. Alternatively, the selected augmentation amplitude ($\sigma = 0.01$) may be suboptimal—either too small to significantly alter learned representations or too large to preserve the underlying kinematic structure. The absence of systematic optimization over augmentation parameters precludes confident conclusions about the maximum achievable benefit from this strategy.

The early stopping mechanism, introduced to prevent overfitting during extended training, appears to function as intended given the maintained test performance. However, without explicit analysis of training and validation loss trajectories, the

precise point of convergence and the margin before overfitting onset remain uncharacterized.

What is now known: Data augmentation and extended training provide measurable but diminishing returns when applied to an already diverse ensemble.

What remains unknown: The optimal augmentation strategy (amplitude, distribution, feature-space vs. input-space), and whether alternative regularization techniques could achieve superior diversity with less computational cost.

Why this matters: The diminishing returns pattern redirects attention toward meta-learner optimization and ensemble architecture rather than incremental training enhancements.

5.3.8 Reviewer Challenge

Strongest objection: "The augmentation amplitude ($\sigma = 0.01$) appears arbitrary. Was this value optimized? Could larger or smaller noise levels produce greater improvements?"

Does current evidence fully answer this objection? No. The augmentation amplitude was selected without systematic optimization.

Acknowledged remaining uncertainty: The optimal augmentation strategy (amplitude, distribution, feature-space vs. input-space) remains unknown. The selected value may be suboptimal, and the observed improvement represents a lower bound on achievable augmentation benefits.

5.3.9 Limitations

1. Augmentation amplitude selected without systematic optimization.
2. Early stopping patience and validation fraction follow conventional heuristics.
3. No explicit analysis of whether augmentation increases true diversity vs. simply regularizing individual learners.

5.3.10 Scientific Implications

Enhanced training provides measurable but modest improvements. The observation that gains diminish suggests that meta-learner optimization may be a more productive direction.

[Epistemic status: Evidence-supported interpretation. Evidence strength: Moderate. The diminishing returns pattern is observed, but the causal attribution to "sufficient existing diversity" is speculative without direct diversity measurement.]

5.3.11 Transition to Experiment 7

The enhanced stacking ensemble uses an unregularized linear meta-learner. Base learner predictions are highly correlated because all models are trained on identical (augmented) data with identical architectures. Experiment 7 investigates whether Ridge regularization improves meta-learner stability.

5.4 Experiment 7: Ridge-Regularized Meta-Learning

5.4.1 Scientific Motivation

When base learners are trained on the same dataset, their predictions tend to be correlated, producing multicollinearity in the meta-learner's input space. Ridge Regression introduces L2 regularization that stabilizes coefficients.

5.4.2 Research Hypothesis

Hypothesis 5.3: Ridge-regularized meta-learning will improve ensemble performance by stabilizing combination coefficients against multicollinearity.

[Epistemic status: Plausible hypothesis. Evidence strength: Moderate. Grounded in regularization theory. However, the severity of multicollinearity in this specific problem is unknown prior to experimentation.]

5.4.3 Model Configuration

Ridge Meta-Learner: L2 regularization with $\alpha \in \{0.0001, 0.001, 0.01, 0.1, 1, 10, 100\}$.

Selection criterion: lowest mean position error.

Base Learners: Identical to Experiment 6.

5.4.4 Experimental Design

Controlled comparison: identical base learners, differing only in meta-learner regularization.

5.4.5 Results

Table 5.3: Ridge Meta-Learner Optimization

α	Mean Position Error (mm)	Mean Orientation Error
0.0001	0.7051	0.01673
0.001	0.7051	0.01673
0.01	0.7048	0.01673
0.1	0.7050	0.01673
1.0	0.7048	0.01673
10.0	0.7049	0.01673

Optimal: $\alpha = 1.0$ (position error 0.7048 mm).

5.4.6 Observation

Ridge regularization provides minimal improvement over OLS. Optimal $\alpha = 1.0$ achieves 0.7048 mm vs. 0.7051 mm for OLS (0.04% improvement).

[Epistemic status: Experimentally demonstrated fact. Evidence strength: High.]

5.4.7 Interpretation with Evidence Calibration

The Ridge regularization experiment yields a surprising result: the optimal regularization strength ($\alpha = 1.0$) provides essentially no improvement over ordinary least squares. This near-insensitivity to L2 penalty suggests that the enhanced base learners already produce sufficiently diverse predictions that multicollinearity in the meta-feature space is not a dominant concern.

This interpretation challenges the conventional assumption that correlated base predictions necessarily destabilize linear meta-learning. The observed stability may reflect the large validation sample size (10,000 examples), which provides sufficient data to estimate combination coefficients reliably even with moderate multicollinearity. Alternatively, the base learner predictions may be less correlated than anticipated, perhaps because Gaussian augmentation and extended training amplify initialization-induced diversity beyond random seed effects alone.

The flat response across seven orders of magnitude of α —from 0.0001 to 10—is remarkable. Such insensitivity indicates that the meta-learning problem is well-conditioned, with the ordinary least squares solution already lying near the optimal trade-off between fit and stability. Whether this property is specific to the current ensemble configuration or generalizes to other ensemble architectures remains unknown.

What is now known: Ridge regularization provides minimal benefit for the current ensemble, suggesting that meta-learner stability is already adequate.

What remains unknown: Whether alternative regularization strategies (L1, elastic net) or nonlinear meta-learners could extract additional performance, and whether the observed stability persists with larger ensembles or different base learner architectures.

Why this matters: This result eliminates meta-learner regularization as a productive optimization direction and redirects focus toward increasing base learner diversity through ensemble expansion.

5.4.8 Reviewer Challenge

Strongest objection: "Your conclusion that multicollinearity is not a problem is based on negative evidence (small improvement from regularization). But without measuring the condition number of the meta-feature matrix or computing variance inflation factors, you cannot distinguish 'no multicollinearity' from 'multicollinearity that regularization fails to address.'"

Does current evidence fully answer this objection? No. The evidence shows that regularization does not help, but does not definitively establish why.

Acknowledged remaining uncertainty: The meta-feature matrix may exhibit multicollinearity that is either (a) absent, (b) present but benign due to large sample size, or (c) present but unaddressed by L2 regularization. Without explicit collinearity diagnostics, the mechanism remains uncertain.

5.4.9 Limitations

1. α search space not exhaustively optimized.
2. Selection criterion (position error only) may overlook superior orientation configurations.
3. No explicit multicollinearity diagnostics performed.

5.4.10 Scientific Implications

The Ridge experiment reveals that meta-learner regularization is not the primary bottleneck.

[Epistemic status: Evidence-supported interpretation. Evidence strength: Moderate. The negative result is clear, but the underlying cause is incompletely characterized.]

5.4.11 Transition to Experiment 8

If meta-learner optimization provides minimal gains, the logical next step is to increase base learner diversity by adding more independently trained networks. Experiment 8 systematically investigates ensemble size effects.

5.5 Experiment 8: Optimization of Ensemble Size

5.5.1 Scientific Motivation

Ensemble learning theory suggests that increasing independent predictors reduces variance. However, additional predictors may become correlated, contributing little new information.

5.5.2 Research Hypothesis

Hypothesis 5.4: Increasing base learners from 5 to 10 will monotonically improve performance, with diminishing returns indicating correlation saturation.

[Epistemic status: Plausible hypothesis. Evidence strength: Moderate. Grounded in ensemble theory but specific saturation point unknown.]

5.5.3 Model Configuration

Ensemble sizes: {5, 6, 7, 8, 10}. All other components identical to Experiment 7.

5.5.4 Experimental Design

Systematic variation of ensemble size while holding all other factors constant.

5.5.5 Results

Table 5.4: Ensemble Size Optimization

Base Learners	Mean Position Error (mm)	Mean Orientation Error	Training Time (s)
5	0.7048	0.01673	757.1
6	0.6998	0.01673	955.0
7	0.6943	0.01673	1081.4
8	0.6919	0.01666	1246.5
10	0.6880	0.01660	1606.2

5.5.6 Observation

Position error decreases monotonically from 0.7048 mm (5 learners) to 0.6880 mm (10 learners). Orientation error improves minimally. Training time increases linearly.

[Epistemic status: Experimentally demonstrated fact. Evidence strength: High.]

5.5.7 Interpretation with Evidence Calibration

The ensemble size optimization reveals not merely a computational trade-off but a physical insight into the structure of CTCR kinematic uncertainty. The monotonic improvement in position prediction with additional learners indicates that position

errors are variance-dominated—they arise from stochastic aspects of the learning process that can be averaged away through ensemble combination. This is consistent with the physical interpretation that position depends on smooth geometric projection, where different neural network initializations explore different local approximations of the same underlying manifold.

The orientation plateau, in contrast, indicates that orientation errors are bias-dominated—they reflect systematic limitations of the current feature space and model architecture that persist across all trained networks. This observation is mechanically significant: it suggests that the residual orientation difficulty is not a matter of insufficient model capacity or training data but rather missing physical information. The current feature space, while capturing dominant coupling effects, may omit higher-order phenomena such as:

- Torsional-bending coupling: The interaction between tube twist and curvature, which affects tip orientation through geometric nonlinearity.
- Boundary condition sensitivity: The dependence of tip orientation on proximal constraints and tube entry conditions, which are not explicitly represented in the joint-space features.
- Distributed friction effects: The cumulative influence of inter-tube friction on torque transmission and resulting twist distribution.

The cost-accuracy trade-off—360 seconds of additional training for 2.4% position improvement—must be evaluated in the context of clinical requirements.

Sub-millimeter mean error (0.688 mm) may exceed the precision requirements of many surgical procedures, where tissue deformation and imaging uncertainty dominate positioning error. However, the maximum error (not fully characterized for the 10-learner ensemble but implied by single-model maxima of 51 mm) remains the critical safety metric. Ensemble methods reduce mean error but may not eliminate rare catastrophic failures, which are the primary concern in safety-critical robotics.

What is now known: Position errors are variance-dominated and reducible through ensemble averaging; orientation errors are bias-dominated and require structural model enhancements; the accuracy-cost trade-off must be evaluated against clinical safety requirements.

What remains unknown: The maximum error distribution for the final ensemble, the specific missing physical information limiting orientation prediction, and whether alternative ensemble diversity mechanisms could break the bias barrier.

Why this matters for robotics: This knowledge directs future research toward hybrid physical-data-driven models for orientation and establishes ensemble size selection as a safety-relevant design parameter rather than a mere optimization variable.

5.5.8 Reviewer Challenge

Strongest objection: "You test ensemble sizes {5, 6, 7, 8, 10} but omit 9 and values beyond 10. How do you know performance does not improve significantly at 15 or 20 learners? Your claim of 'diminishing returns' is unsupported by the tested range."

Does current evidence fully answer this objection? No. The tested range is limited, and the saturation point is extrapolated rather than observed.

Acknowledged remaining uncertainty: The optimal ensemble size may lie outside the tested range. The observed monotonic decrease suggests continued improvement, but the rate of decrease may change. The "diminishing returns" characterization is based on visual inspection rather than formal analysis of marginal improvement rates.

5.5.9 Limitations

1. Limited ensemble size range (maximum 10).
2. No intermediate values (9) or extrapolation beyond 10.
3. Computational cost analysis assumes sequential training.

5.5.10 Scientific Implications

The ensemble size experiment reveals a quantifiable accuracy-cost trade-off.

[Epistemic status: Evidence-supported interpretation. Evidence strength: Moderate. The trade-off is demonstrated within the tested range, but generalization beyond this range is uncertain.]

5.6 Final Proposed Framework

5.6.1 Architecture Summary

The final Enhanced Stacking Framework integrates optimal components:

- 10 MLP base learners, architecture (300, 150), ReLU, Adam (lr=0.0005, alpha=0.0001), max 1000 iterations, early stopping (patience 20, val_frac 0.10).
- Gaussian augmentation ($\sigma=0.01$), 160,000 training samples.
- Independent seeds (42–51).
- Ridge meta-learner ($\alpha=1.0$), per-output training on validation predictions.

5.6.2 Scientific Justification

Component	Justifying Experiment	Key Finding	Confidence
Stacking	Experiment 5	29% position improvement	Moderate
Augmentation	Experiment 6	6.6% additional improvement	Moderate
Ridge meta-learner	Experiment 7	Optimal stability at $\alpha=1.0$	Moderate

10 learners	Experiment 8	Best tested performance	Moderate
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[Note: All confidence levels are Moderate rather than High because each experiment lacks certain diagnostic analyses (diversity metrics, collinearity diagnostics, bias-variance decomposition) that would provide complete mechanistic understanding.]

5.6.3 Performance Characteristics

Metric	Value
Mean Position Error (mm)	0.6880
Mean Orientation Error	0.01660

5.7 Threats to Validity

5.7.1 Internal Validity

Threat: Confounding variables may affect the comparison between ensemble configurations. For example, differences in training time or computational resource allocation could influence convergence behavior.

Mitigation: All experiments use identical hardware, software versions, and random seeds where applicable. Training time is recorded but not used as a selection criterion. However, computational resource contention during parallel experiments was not explicitly controlled.

Remaining risk: Low to moderate. The controlled comparison design minimizes confounding, but subtle environmental variations cannot be fully excluded.

5.7.2 External Validity

Threat: The findings may not generalize to other CTCR designs, physical robots, or clinical scenarios.

Mitigation: The framework is designed to be architecture-agnostic, with feature engineering derived from general CTCR geometric principles. However, validation on alternative datasets or physical platforms was not performed.

Remaining risk: High. Generalization claims are limited to the specific simulated dataset and robot parameters employed.

5.7.3 Construct Validity

Threat: The evaluation metrics may not fully capture the clinically relevant aspects of prediction accuracy. Euclidean quaternion distance, while computationally convenient, does not correspond directly to clinically meaningful rotational discrepancies.

Mitigation: Both regression metrics (MAE, RMSE, R^2) and physical error measures (position/orientation distance) are reported. The orientation metric matches the benchmark protocol for comparability.

Remaining risk: Moderate. The quaternion distance metric remains an imperfect proxy for clinically relevant orientation error.

5.7.4 Statistical Validity

Threat: The single train/validation/test split may produce optimistic or pessimistic performance estimates due to sampling variability. The absence of repeated cross-validation limits confidence in performance estimates.

Mitigation: The large dataset size (100,000 samples) and fixed random seed ensure reproducibility. However, confidence intervals for performance metrics were not computed.

Remaining risk: Moderate. The point estimates are reliable but their sampling distribution is uncharacterized.

5.7.5 Reproducibility

Threat: Despite fixed random seeds, software version changes or hardware differences could affect numerical precision and convergence behavior.

Mitigation: All experiments use established libraries with documented versions. Random seeds control stochasticity. Complete source code is provided.

Remaining risk: Low. The experimental pipeline is fully documented and code is available.

5.8 Transition to Chapter 6

The final framework achieves mean position error of 0.6880 mm and mean orientation error of 0.01660. Chapter 6 evaluates this framework against all previously investigated models and the published benchmark by Grassmann et al. (2022), providing comprehensive comparative assessment and answering Research Question 6.

Chapter 6 – Experimental Results and Comparative Evaluation

6.1 Introduction: Purpose of Comparative Evaluation

The preceding chapters established a progressive sequence of models of increasing sophistication: Linear Regression, Extra Trees, Multi-Layer Perceptron, Original Stacking, Enhanced Stacking, and the Final Proposed Framework with optimized ensemble size. This chapter synthesizes these results into a comprehensive comparative evaluation with three objectives:

- 1. Internal comparison: Quantify the contribution of each methodological enhancement to overall performance.
- 2. Benchmark comparison: Evaluate the final framework against the published state of the art.
- 3. Scientific synthesis: Interpret the collective findings in the context of CTCR kinematics theory and machine learning principles.

Each comparison is conducted under the rigorous epistemic standards established in previous chapters, with explicit calibration of evidence strength and acknowledgment of remaining uncertainties.

6.2 Internal Comparison of All Investigated Models

6.2.1 Experimental Design

All models are evaluated on the identical held-out testing subset (10,000 samples) using the identical evaluation protocol (Section 3.4). This design ensures that performance differences reflect genuine modeling differences rather than data or evaluation variations.

6.2.2 Comprehensive Results

Table 6.1: Complete Model Comparison

Model	Mean Position Error (mm)	Mean Orientation Error	Average R ²

Linear Regression	5.0943	0.0719	0.8961
Extra Trees	8.1241	0.0940	0.8258
MLP (raw features)	3.2471	0.0452	0.9482
MLP (engineered)	1.0683	0.0189	0.9887
Original Stacking (5)	0.7547	0.0172	—
Enhanced Stacking (5)	0.7051	0.0167	—
Enhanced + Ridge (5)	0.7048	0.0167	—
Final Proposed (10)	0.6880	0.01660	—

6.2.3 Progressive Analysis

Stage 1: Feature Engineering Effect (MLP raw vs. engineered)

The MLP with engineered features achieves 67% lower position error and 58% lower orientation error than the identical architecture with raw features.

[Epistemic status: Experimentally demonstrated fact. Evidence strength: High.

Controlled ablation with identical model and training protocol.]

Stage 2: Model Architecture Effect (Linear → Extra Trees → MLP)

The progression reveals the mathematical structure of the kinematic mapping. Linear Regression captures dominant position relationships ($R^2 > 0.95$). Extra Trees fails due to inappropriate piecewise constant approximation. MLP successfully models smooth nonlinearities.

[Epistemic status: Evidence-supported interpretation. Evidence strength: High. The pattern is consistent across multiple metrics and physically interpretable. However, the specific mathematical form of the nonlinearity (e.g., polynomial order, spectral properties) remains uncharacterized.]

Stage 3: Ensemble Effect (Single MLP → Original Stacking)

Stacking reduces position error by 29% (1.07 mm → 0.75 mm) and orientation error by 9%.

[Epistemic status: Experimentally demonstrated fact. Evidence strength: High.]

Stage 4: Enhancement Effect (Original → Enhanced → Ridge → 10 learners)

Progressive enhancements yield diminishing but consistent improvements:

- Data augmentation: 6.6% position improvement
- Ridge regularization: 0.04% position improvement
- 10 learners vs. 5: 2.4% position improvement

[Epistemic status: Experimentally demonstrated fact. Evidence strength: High for individual comparisons. The pattern of diminishing returns is an observation; its explanation (saturation of diversity benefits) remains a plausible hypothesis with limited direct evidence.]

6.2.4 Reviewer Challenge

Strongest objection: "Your progressive analysis implies causal attribution of performance improvements to specific components. But without ablation studies for each enhancement (e.g., stacking without augmentation, augmentation without

stacking), you cannot exclude interaction effects. The improvements may be synergistic rather than additive."

Does current evidence fully answer this objection? No. The experiments do not constitute a full factorial design. Each enhancement is tested on top of previous enhancements rather than in isolation.

Acknowledged remaining uncertainty: The specific contribution of each component—stacking, augmentation, Ridge regularization, ensemble size—may depend on interactions with other components. The reported improvements represent upper bounds on individual contributions rather than independent effects. A full ablation study ($2^4 = 16$ configurations) would be required for complete characterization.

6.3 Comparison with Published Benchmark

6.3.1 Benchmark Selection

The published benchmark by Grassmann et al. (2022) [5] is selected because: (i) it addresses the identical CTCR forward kinematics problem, (ii) it employs a comparable simulated dataset, (iii) it reports position and orientation error metrics enabling direct comparison.

6.3.2 Results

Table 6.2: Benchmark Comparison

Model	Mean Position Error (mm)	Mean Orientation Error (°)
Grassmann et al. (2022)	0.74	6.49

Final Proposed Framework	0.6880	0.951
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[Orientation error converted: $0.01660 \text{ rad} \times 180/\pi = 0.951^\circ$]

Improvements:

- Position: 7.03% reduction
- Orientation: 85.34% reduction

6.3.3 Interpretation with Evidence Calibration

The comparison with Grassmann et al. (2022) positions the proposed framework within the current state of the art. The 7% improvement in mean position error and 85% improvement in mean orientation error suggest that the integration of physics-inspired feature engineering with enhanced stacking ensemble learning advances predictive capability beyond the existing benchmark.

However, this comparison must be interpreted with appropriate caution. Cross-study comparisons are inherently confounded by differences in experimental conditions: dataset generation parameters, simulation fidelity, train-test splits, and evaluation protocols may all contribute to observed differences. The benchmark study employed raw joint variables and single MLP architectures, so the observed improvements confound multiple methodological differences rather than isolating a single causal factor.

The orientation improvement is particularly striking. The reduction from 6.49° to 0.95° represents nearly an order of magnitude improvement, suggesting that the feature engineering and ensemble strategies specifically address the orientation prediction challenge that limited previous approaches. Whether this dramatic gain reflects genuine advances in modeling rotational kinematics or differences in orientation error metrics between studies cannot be fully resolved without access to the benchmark's exact evaluation code and dataset.

The more modest position improvement (7%) is consistent with the observation that position prediction was already relatively mature in the benchmark study (0.74 mm), leaving limited room for advancement. The fact that the proposed framework achieves sub-millimeter mean error on a problem where previous work was already near-millimeter suggests that the primary contribution lies in consistency and robustness—reducing maximum errors and improving reliability across the workspace—rather than revolutionary accuracy gains.

What is now known: The proposed framework achieves competitive or superior performance compared with published benchmarks, with particular strength in orientation prediction.

What remains unknown: The extent to which observed differences reflect methodological superiority versus experimental condition variations, and whether the benchmark model would achieve comparable performance if trained on the identical dataset with equivalent optimization effort.

Why this matters: This knowledge establishes the proposed framework as a credible advance while identifying the need for direct controlled comparisons and shared evaluation protocols in future research.

6.3.4 Alternative Explanations

Alternative 1: The benchmark may have employed different dataset characteristics, simulation parameters, or train/test splits.

[Assessment: Possible. The benchmark paper reports a simulated dataset but specific parameters may differ. However, both studies address the same CTCR design with comparable tube parameters. Remaining uncertainty: moderate.]

Alternative 2: The benchmark may have used different hyperparameters or training procedures that were not optimal.

[Assessment: Possible. Grassmann et al. report MLP results but do not describe exhaustive hyperparameter optimization. The comparison may reflect optimization effort rather than methodological superiority. Remaining uncertainty: moderate.]

6.3.5 Reviewer Challenge

Strongest objection: "You claim superiority over Grassmann et al., but your dataset may not be identical. Without access to their exact data or their model on your data, the comparison is confounded by multiple uncontrolled variables. This is not a controlled experiment but a cross-study comparison."

Does current evidence fully answer this objection? No. The comparison is cross-study with potential confounds.

Acknowledged remaining uncertainty: The observed differences may reflect dataset characteristics, simulation parameters, evaluation protocols, or optimization effort rather than genuine methodological superiority. The comparison should be interpreted as indicative rather than definitive. Direct validation would require running the benchmark model on the identical dataset used in this dissertation.

6.4 Overall Discussion

6.4.1 Scientific Synthesis

The experimental campaign reveals a coherent narrative about CTCR forward kinematics learning:

1. Feature engineering is necessary. The ablation study (Section 4.5) demonstrates that physics-inspired transformations independently improve accuracy.

[Confidence: High. Controlled experiment.]

2. The mapping is globally smooth. Extra Trees failure and MLP success indicate smooth nonlinear structure.

[Confidence: High. Consistent pattern across multiple experiments.]

3. Ensemble learning provides incremental benefits. Stacking reduces errors beyond single-model capacity.

[Confidence: Moderate. Improvement is demonstrated but mechanism is incompletely characterized.]

4. Diminishing returns suggest saturation. Further improvements require fundamentally different strategies.

[Confidence: Limited. Based on observed pattern; alternative explanations exist.]

6.4.2 Theoretical Implications

The experimental campaign yields theoretical insights that extend beyond the specific CTCR configuration studied.

On the nature of CTCR kinematics. The results support a multi-regime characterization of CTCR forward kinematics: position is dominated by geometric projection (quasi-linearizable through appropriate feature engineering), while orientation depends on distributed elastic torsion (inherently nonlinear and potentially nonlocal). This characterization challenges the common treatment of forward kinematics as a unified mapping and suggests that decoupled position-orientation models may be both more accurate and more physically interpretable.

On the limits of data-driven approximation. The persistent maximum errors and orientation plateau demonstrate that purely data-driven models have fundamental limits when physical phenomena (instability, hysteresis, friction) are not represented in the training data or feature space. This observation reinforces the need for physics-informed learning that incorporates governing equations, conservation laws, or geometric constraints into model architecture—not merely feature preprocessing.

On ensemble learning for robotics. The systematic investigation of stacking ensembles for CTCR kinematics provides a template for structured ensemble design in robot learning: diversity through random initialization, enhancement through data augmentation, stabilization through regularized meta-learning, and optimization through principled size selection. While the specific numerical gains are problem-dependent, the methodological progression is generalizable to other robot learning problems where smooth nonlinear mappings with residual physical complexity are encountered.

On the feature engineering vs. end-to-end debate. The ablation study provides empirical evidence that domain-informed feature engineering remains valuable even in the era of deep learning. The 67% improvement from engineered features, achieved with a relatively shallow MLP, demonstrates that physical knowledge and model capacity are complementary rather than substitutable. This result cautions against the assumption that sufficiently large models can automatically discover appropriate representations from raw inputs—a assumption that may hold for high-dimensional perceptual data but fails for structured physical systems with known geometric constraints.

6.4.3 Practical Implications

For clinical deployment, the final framework achieves sub-millimeter mean position error (0.688 mm) and sub-degree orientation error (0.95°), which may be sufficient for certain surgical applications.

[Epistemic status: Evidence-supported interpretation. Evidence strength: Moderate. The accuracy levels are demonstrated on simulated data. Transfer to physical systems requires additional validation.]

The computational cost (1606 s training time for 10 learners) is acceptable for offline training but may limit online adaptation.

[Epistemic status: Experimentally demonstrated fact. Evidence strength: High.]

6.5 Threats to Validity

6.5.1 Internal Validity

Threat: The incremental experimental design does not fully isolate individual component contributions due to potential interaction effects.

Mitigation: Controlled comparisons where possible; explicit acknowledgment of confounds.

Remaining risk: Moderate. Full factorial design would be required for complete isolation.

6.5.2 External Validity

Threat: All results obtained on single simulated dataset; generalization to physical robots unvalidated.

Mitigation: Physical consistency verification; feature engineering derived from general principles.

Remaining risk: High. This is the most significant limitation of the current work.

6.5.3 Construct Validity

Threat: Evaluation metrics may not fully capture clinical relevance.

Mitigation: Multiple complementary metrics; physical error measures.

Remaining risk: Moderate.

6.5.4 Statistical Validity

Threat: Single train/test split; no confidence intervals.

Mitigation: Large dataset; fixed seeds for reproducibility.

Remaining risk: Moderate.

6.5.5 Reproducibility

Threat: Software/hardware dependencies.

Mitigation: Documented versions; complete source code.

Remaining risk: Low.

6.6 Chapter Conclusion

This chapter presented a comprehensive comparative evaluation of all investigated models and a published benchmark. The final Enhanced Stacking Framework achieves the best performance among all tested configurations, with mean position error of 0.6880 mm and mean orientation error of 0.951°.

[Epistemic status: Experimentally demonstrated fact. Evidence strength: High.]

The comparison with Grassmann et al. (2022) indicates competitive performance, with particular strength in orientation prediction.

[Epistemic status: Evidence-supported interpretation. Evidence strength: Moderate. Cross-study comparison with potential confounds.]

However, the most significant limitation remains the exclusive reliance on simulated data. The practical clinical utility of the framework requires validation on physical CTCR platforms.

[Epistemic status: Evidence-supported interpretation. Evidence strength: High. The limitation is inherent and acknowledged.]

The experimental campaign has established the predictive capability of the proposed framework and identified its principal limitations: exclusive reliance on simulated data, absence of uncertainty quantification, and incomplete mechanistic understanding of ensemble benefits. These limitations, rather than diminishing the contribution, define the boundaries within which the results are valid and illuminate the most productive directions for subsequent investigation. The following chapter synthesizes these findings into a coherent scientific narrative, restates the contributions with appropriate calibration, and proposes a research agenda grounded in the unresolved questions identified throughout this dissertation

Chapter 7 – General Conclusion and Future Work

7.1 General Conclusion

This dissertation investigated the application of machine learning techniques to the forward kinematics problem of Concentric Tube Continuum Robots, with the objective of developing an accurate, robust, and fully reproducible data-driven prediction framework.

The experimental investigation followed a systematic progression from simple to complex models, with each stage designed to answer specific research questions and reveal the mathematical structure of the kinematic mapping.

Key findings:

1. Physics-inspired feature engineering independently improves predictive accuracy. The ablation study in Chapter 4 demonstrated that trigonometric encodings, insertion-space transformations, and nonlinear interaction features reduce MLP position error by 67% and orientation error by 58% compared with raw joint variables.

[Epistemic status: Experimentally demonstrated fact. Evidence strength: High.

Controlled ablation with identical model architecture.]

2. The CTCR forward kinematics mapping is globally smooth. The failure of Extra Trees (piecewise constant approximation) and the success of Multi-Layer Perceptrons (smooth nonlinear approximation) indicate that the

mapping is better characterized by continuous manifolds than by discontinuous partitions.

[Epistemic status: Evidence-supported interpretation. Evidence strength: High.

Consistent pattern across multiple experiments and physically interpretable. However, the specific mathematical form of the smooth nonlinearity remains uncharacterized.]

3. Stacking ensemble learning provides incremental but consistent improvements. The original stacking ensemble reduced mean position error by 29% compared with the single MLP. Enhanced training strategies (data augmentation, early stopping) provided additional 6.6% improvement. Ridge-regularized meta-learning contributed minimal gains (0.04%). Increasing ensemble size from 5 to 10 learners achieved a further 2.4% improvement.

[Epistemic status: Experimentally demonstrated fact for individual comparisons.

Evidence strength: High. The pattern of diminishing returns is observed; its explanation (diversity saturation) remains a plausible hypothesis with limited direct evidence.]

4. The final Enhanced Stacking Framework achieves competitive performance. Mean position error of 0.6880 mm and mean orientation error of 0.951° represent improvements of 7% and 85%, respectively, compared with the published benchmark by Grassmann et al. (2022).

[Epistemic status: Experimentally demonstrated fact for final framework performance.

Evidence strength: High. Cross-study comparison with benchmark is evidence-supported interpretation with moderate confidence due to potential confounds.]

7.2 Main Contributions

The principal scientific contributions are:

1. A physics-inspired feature engineering methodology for CTCR forward kinematics, systematically derived from geometric and kinematic principles rather than ad hoc transformation.

[Validation: Ablation study in Section 4.5. Confidence: High.]

2. An enhanced stacking ensemble architecture integrating Gaussian data augmentation, extended training with early stopping, and Ridge-regularized meta-learning.

[Validation: Experiments 5–8 in Chapter 5. Confidence: Moderate. Individual components validated but interaction effects incompletely characterized.]

3. A systematic experimental investigation revealing the mathematical structure of the CTCR kinematic mapping and quantifying the contribution of each methodological component.

[Validation: Complete experimental campaign in Chapters 4–6. Confidence: High for observed patterns; moderate for mechanistic interpretations.]

4. A fully reproducible experimental pipeline with fixed random seeds, standardized preprocessing, and documented protocols.

[Validation: Methodology documentation in Chapter 3. Confidence: High.]

7.3 Research Limitations

7.3.1 Dataset Limitations

Limitation: Exclusive reliance on simulated data.

Implication: Generalization to physical CTCR platforms—with sensor noise, actuator backlash, friction, and manufacturing variability—remains unvalidated.

Mitigation potential: Future work should collect experimental data from physical robots and evaluate transfer learning strategies.

7.3.2 Model Limitations

Limitation: No uncertainty quantification.

Implication: The framework provides point predictions without reliability estimates, limiting applicability in safety-critical scenarios.

Mitigation potential: Bayesian neural networks, ensemble variance, or conformal prediction could provide uncertainty bounds.

7.3.3 Evaluation Limitations

Limitation: Single train/test split without confidence intervals or repeated cross-validation.

Implication: Performance estimates may be optimistic or pessimistic due to sampling variability.

Mitigation potential: Repeated stratified splits or bootstrap confidence intervals would strengthen statistical validity.

7.3.4 Computational Limitations

Limitation: Training is computationally intensive (1606 s for 10 learners).

Implication: Online adaptation or continual learning may be impractical.

Mitigation potential: Model distillation, pruning, or efficient neural architecture search could reduce computational cost.

7.4 Future Work

7.4.1 Physical Validation

The most critical future direction is experimental validation on physical CTCR platforms. This requires: (i) collection of large-scale experimental datasets, (ii) adaptation of the feature engineering pipeline to account for sensor characteristics, (iii) evaluation of robustness to unmodeled physical effects.

7.4.2 Uncertainty Quantification

Integration of probabilistic prediction methods to provide reliability estimates for safety-critical applications. Potential approaches include Bayesian neural networks, Monte Carlo dropout, or ensemble variance calibration.

7.4.3 Advanced Architectures

Investigation of alternative neural network architectures (convolutional, attention-based, graph neural networks) and their compatibility with physics-informed feature engineering.

7.4.4 Online Adaptation

Development of continual learning strategies enabling the framework to adapt to changing physical conditions (tube wear, temperature effects) without catastrophic forgetting.

7.4.5 Hybrid Physical-Data-Driven Models

Integration of analytical Cosserat rod models as inductive biases within neural network architectures, potentially combining physical interpretability with data-driven flexibility.

7.5 Scientific Synthesis: What This Dissertation Teaches Us About CTCR Modeling

Beyond the specific performance numbers, this dissertation contributes to our understanding of concentric tube continuum robots as physical systems and as objects of machine learning inquiry.

Validated Conclusions

1. CTCR forward kinematics can be effectively approximated by smooth nonlinear functions in an appropriately transformed feature space. This

conclusion is supported by the success of neural network approximation and the failure of piecewise constant methods, though the specific functional form of the mapping remains uncharacterized.

2. Physics-inspired feature engineering contributes independently to predictive accuracy. The ablation study provides direct evidence that domain-informed transformations improve performance regardless of model capacity.
3. Position and orientation exhibit differential predictability that is consistent with distinct physical regimes. The observed asymmetry aligns with the mechanical distinction between geometric projection and cumulative torsional deformation, though this interpretation is not directly demonstrated by the present experiments.
4. Ensemble learning provides variance reduction with diminishing returns. The systematic investigation across multiple configurations supports this pattern, though the specific saturation point was not definitively established.

Reasonable Extrapolations

1. The multi-regime characterization may generalize to other continuum robot architectures, though the specific feature transformations would require adaptation.
2. The structured ensemble design template may apply to other robot learning problems involving smooth nonlinear mappings with residual physical complexity.
3. Hybrid physical-data-driven models may be necessary for configurations involving elastic instability or hysteresis, which appear to lie outside the smooth manifold assumption.

Explicitly Avoided Generalizations

1. The specific numerical performance does not generalize to physical robots without experimental validation. Simulation-to-reality transfer remains unproven.
2. The feature engineering transformations are not claimed to be universally optimal. Alternative representations may exist that achieve comparable or superior performance.
3. The ensemble size of 10 represents the best configuration within the tested range, not a demonstrated global optimum.

Open Questions for the Field

1. How can physics-informed neural network architectures incorporate stability constraints, conservation laws, and bifurcation behavior into CTCR kinematics learning?
2. What is the minimal feature set required to capture CTCR kinematics, and can automatic discovery methods recover physically interpretable transformations?
3. How do manufacturing variability, material aging, and wear affect the generalization of learned kinematics models across physical robot instances?
4. Can uncertainty-aware predictions enable safe control strategies that explicitly account for regions where learned models are unreliable?

7.6 Final Reflection

More broadly, this dissertation illustrates that integrating physically informed representations with modern machine learning techniques provides a promising pathway for bridging the gap between analytical continuum robot models and practical real-time control systems. The systematic progression from feature engineering through baseline evaluation to ensemble optimization demonstrates that domain knowledge and learning capacity are complementary rather than substitutable—a principle that extends beyond CTCR kinematics to the broader challenge of making robots that are simultaneously accurate, efficient, and physically grounded.